

WILO CRONOLINE IL SERIES INSTALLATION AND OPERATING INSTRUCTIONS MANUAL



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Fig. 1,

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Wilo-CronoLine-IL **Wilo-CronoBloc-BL**

(mit Kartuschen-Gleitringdichtung/with cartridge mechanical seal/avec garniture mécanique à cartouche/met mechanische asafdichting met cartridge)



- de** Einbau- und Betriebsanleitung
- en** Installation and operating instructions
- fr** Notice de montage et de mise en service
- nl** Inbouw- en bedieningsvoorschriften

Demontage der Gleitringdichtung / Unmounting the mechanical seal /
Démontage de la garniture mécanique / Demontage van de mechanische afdichting

Fig. 1

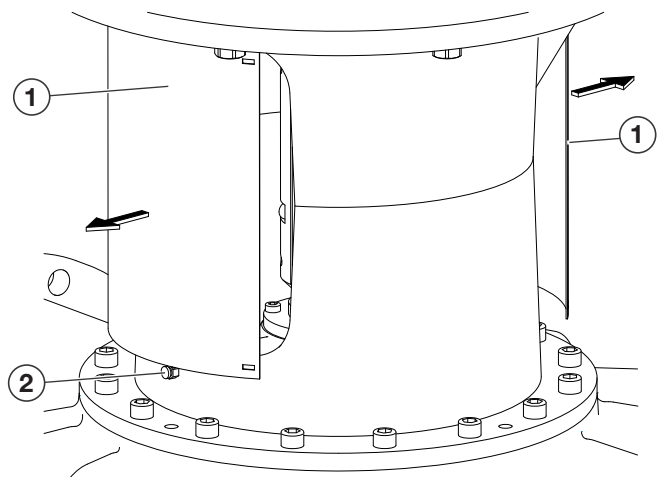


Fig. 2

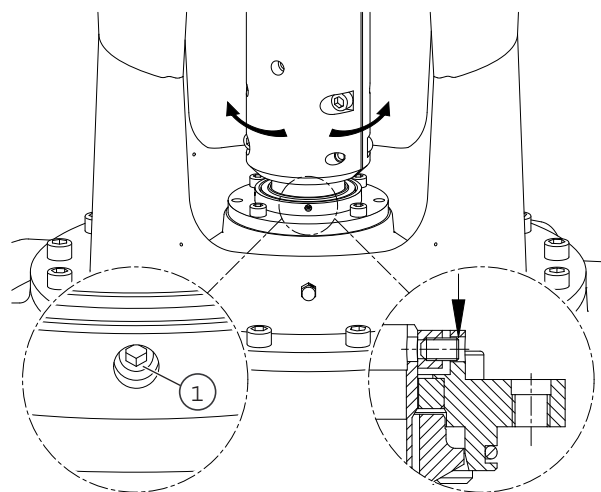


Fig. 3

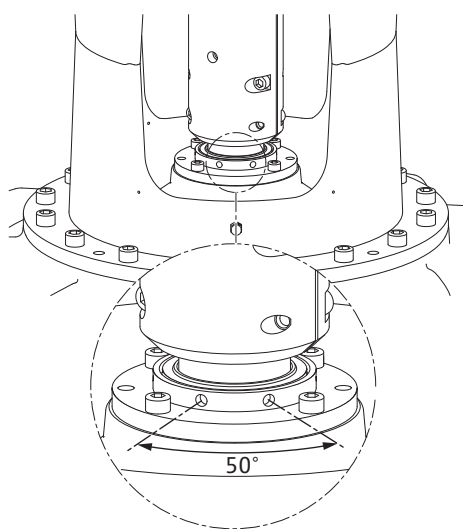


Fig. 4

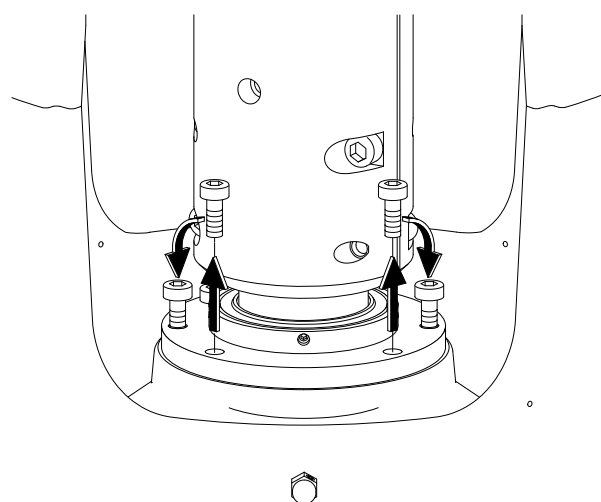


Fig. 5

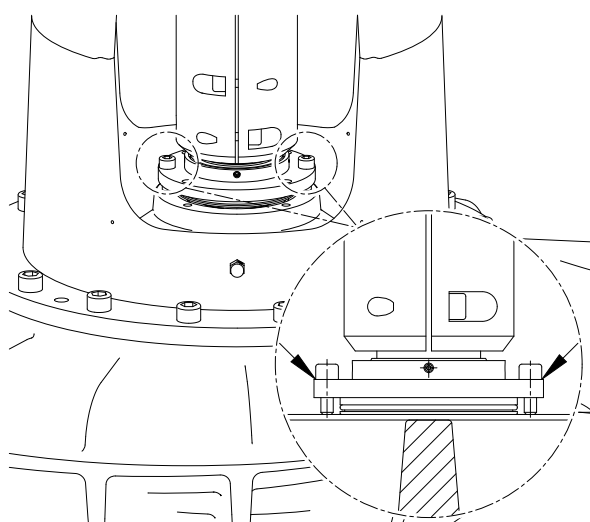


Fig. 6

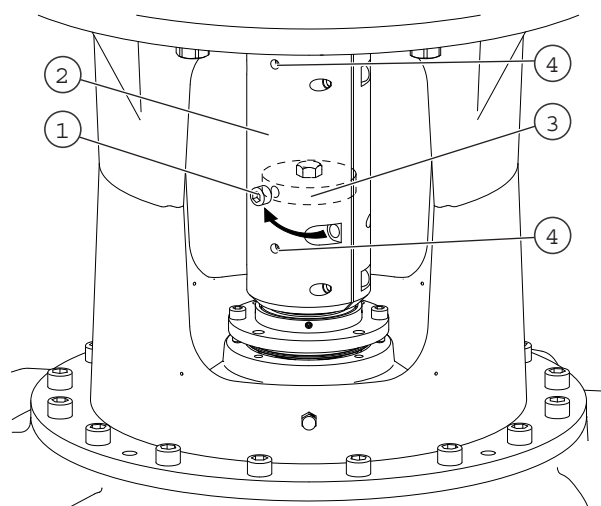


Fig. 7

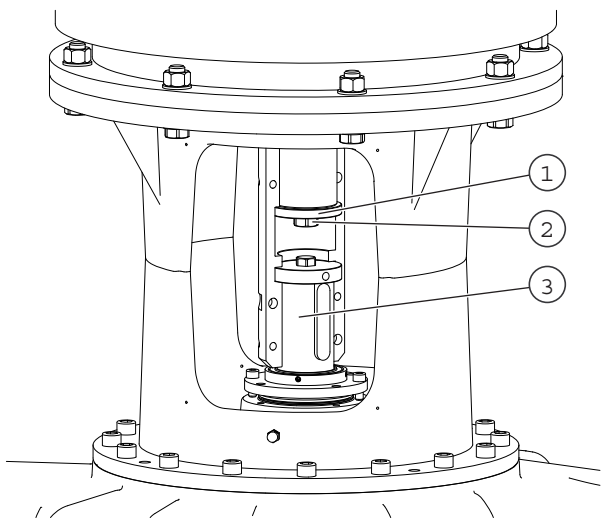


Fig. 8

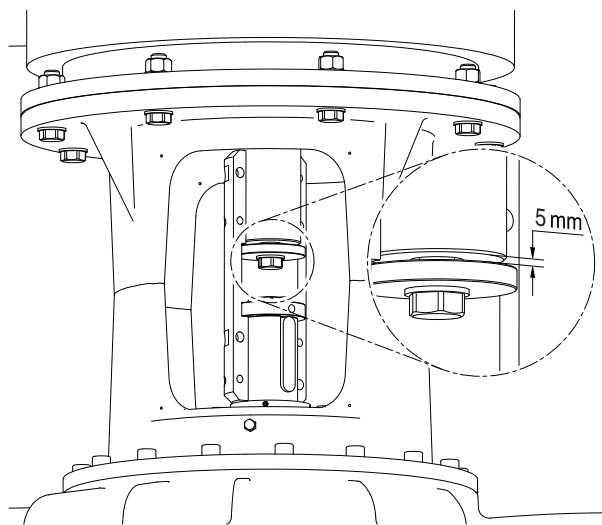


Fig. 9

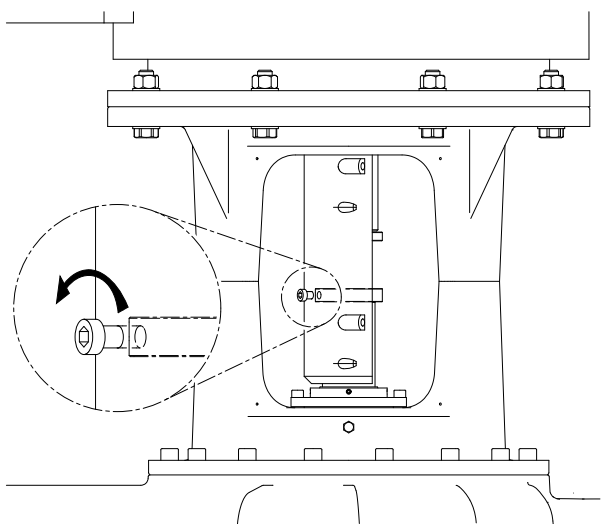


Fig. 10

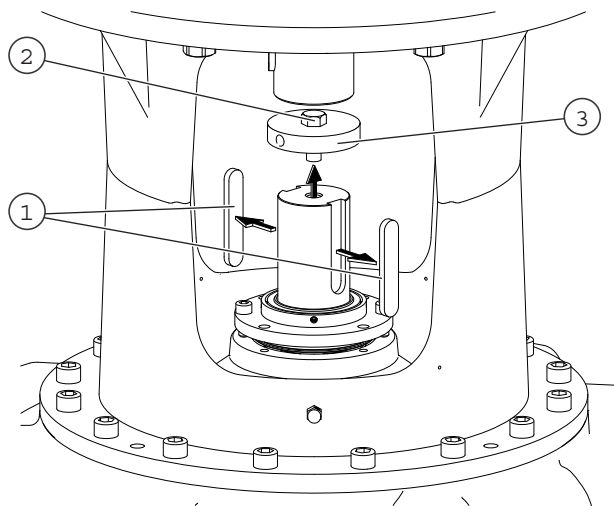
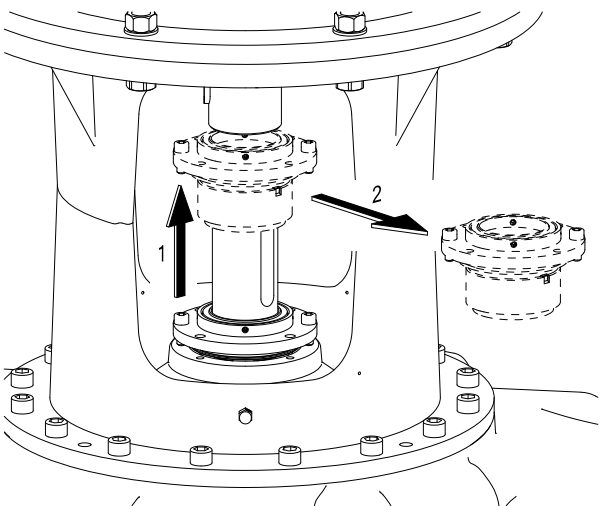


Fig. 11



Montage der Gleitringdichtung / Mounting the mechanical seal /
Montage de la garniture mécanique / Montage van de mechanische
afdichting

Fig. 12

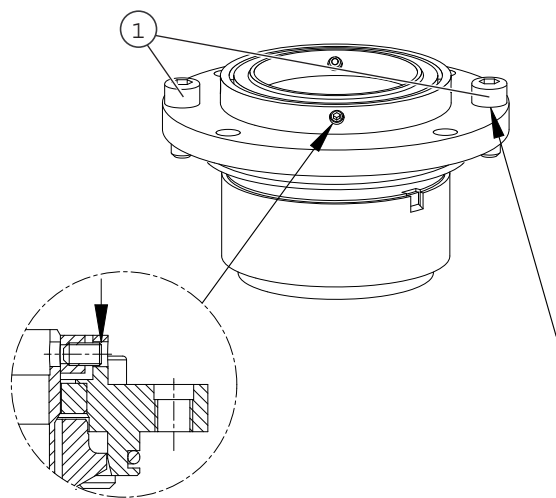


Fig. 13

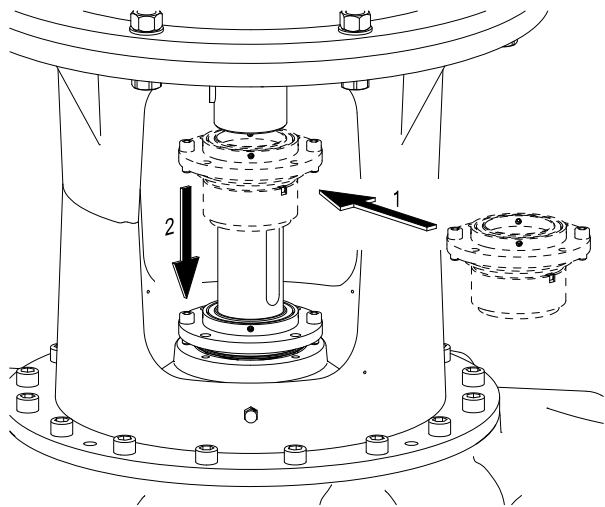


Fig. 14

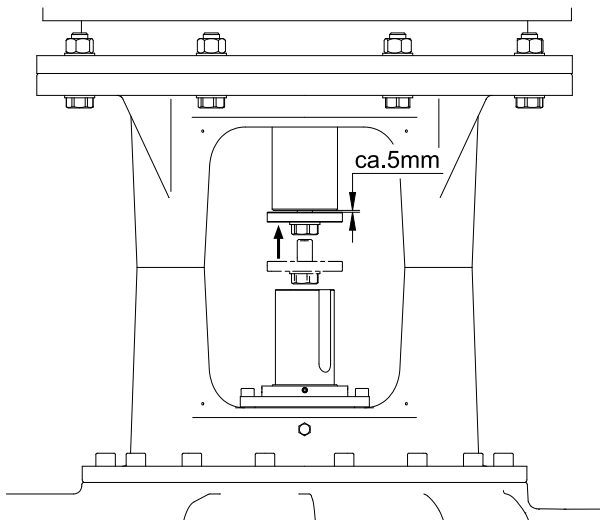


Fig. 15

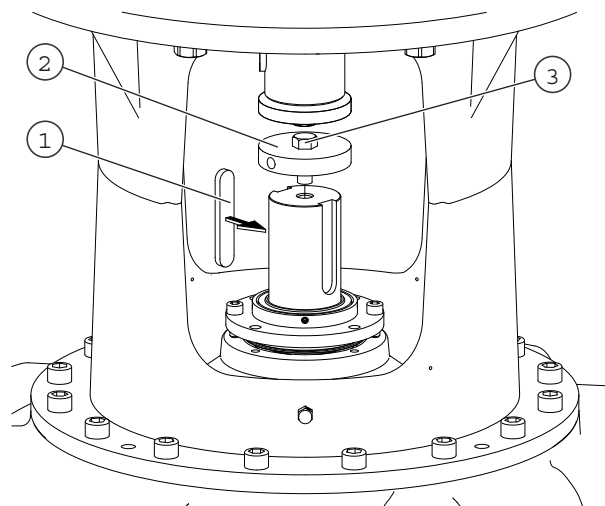


Fig. 16

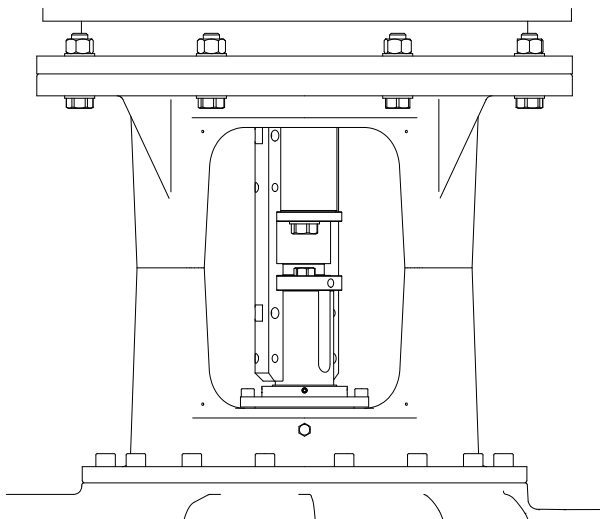


Fig. 17

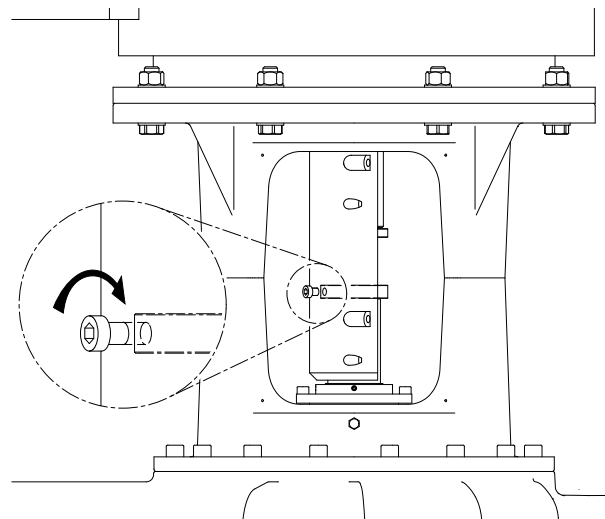


Fig. 18

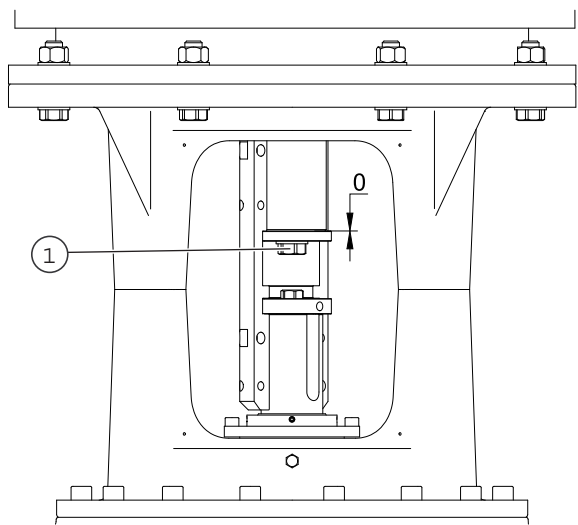


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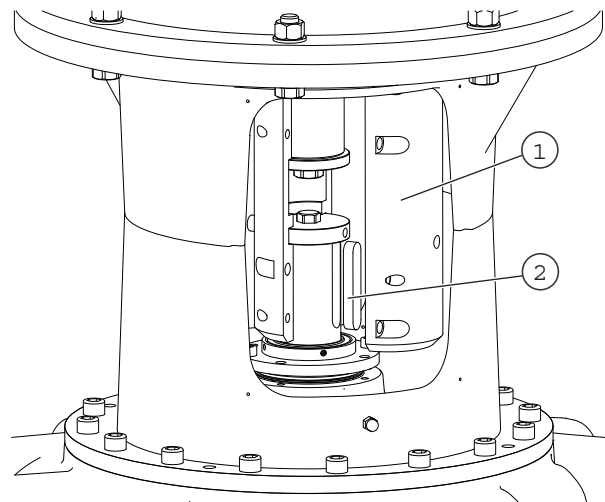


Fig. 20

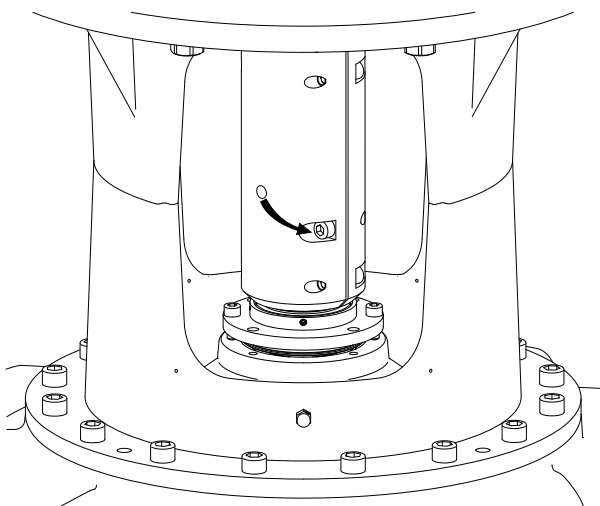


Fig. 21

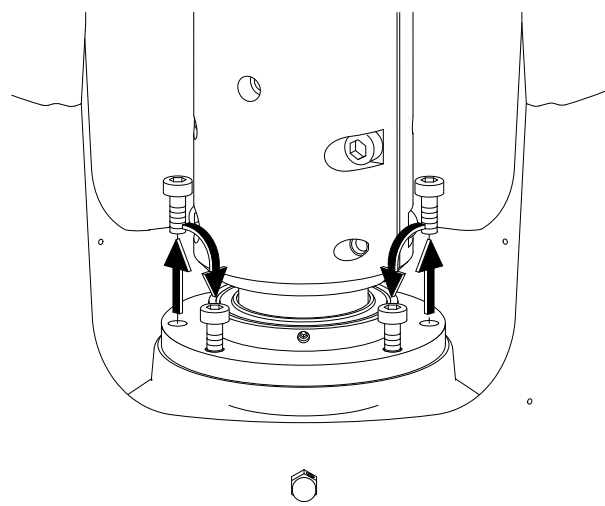


Fig. 22

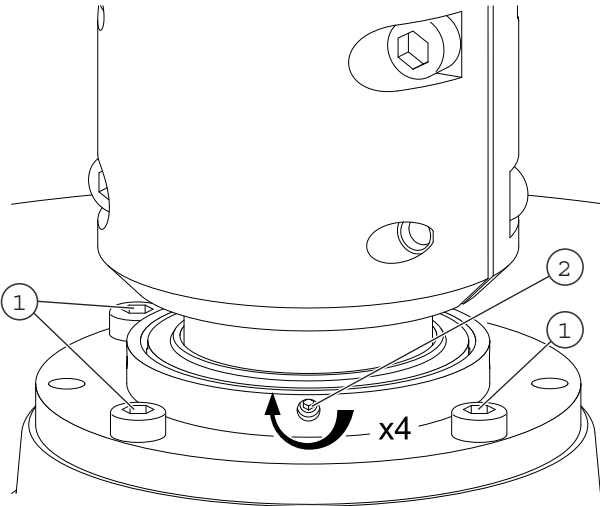
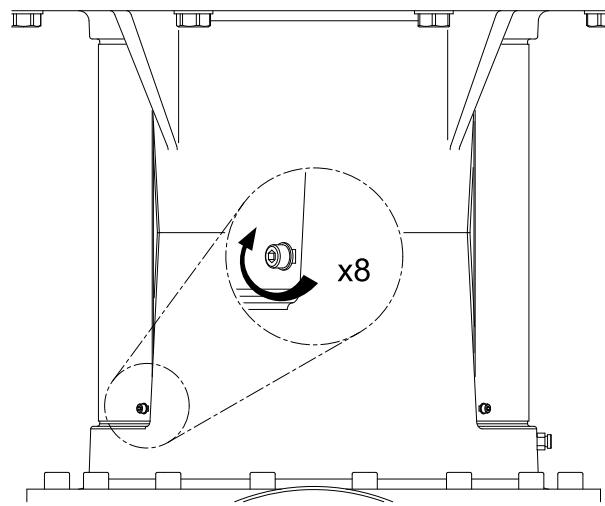


Fig. 23



Motorwechsel / Exchange of Motor / Remplacement du moteur /
Vervanging van de motor

Fig. 24

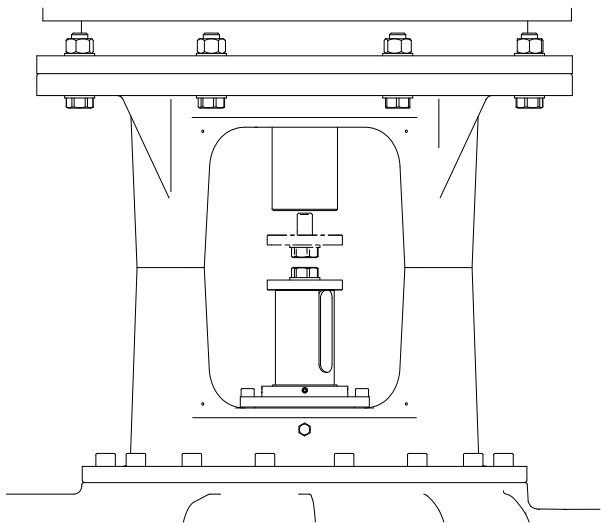


Fig. 25

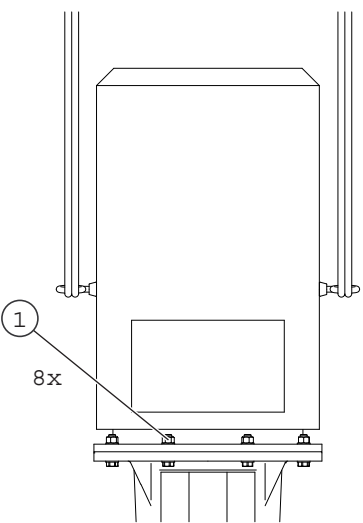


Fig. 26

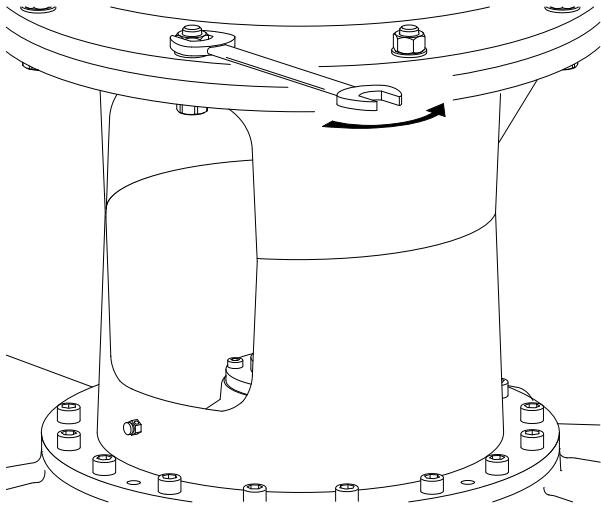


Fig. 27

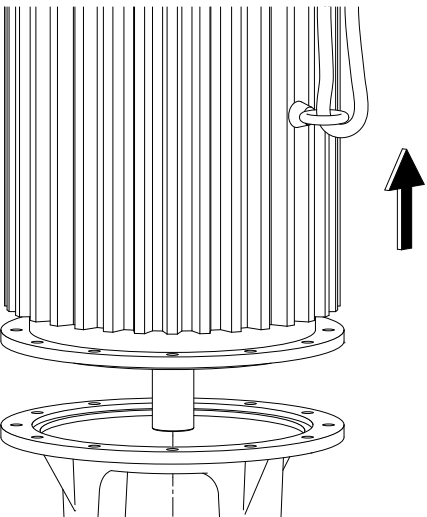


Fig. 28

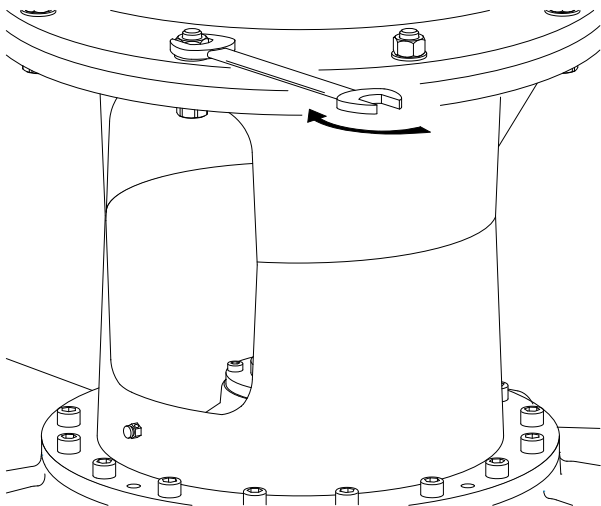


Fig. 29: IL Back Pull-Out

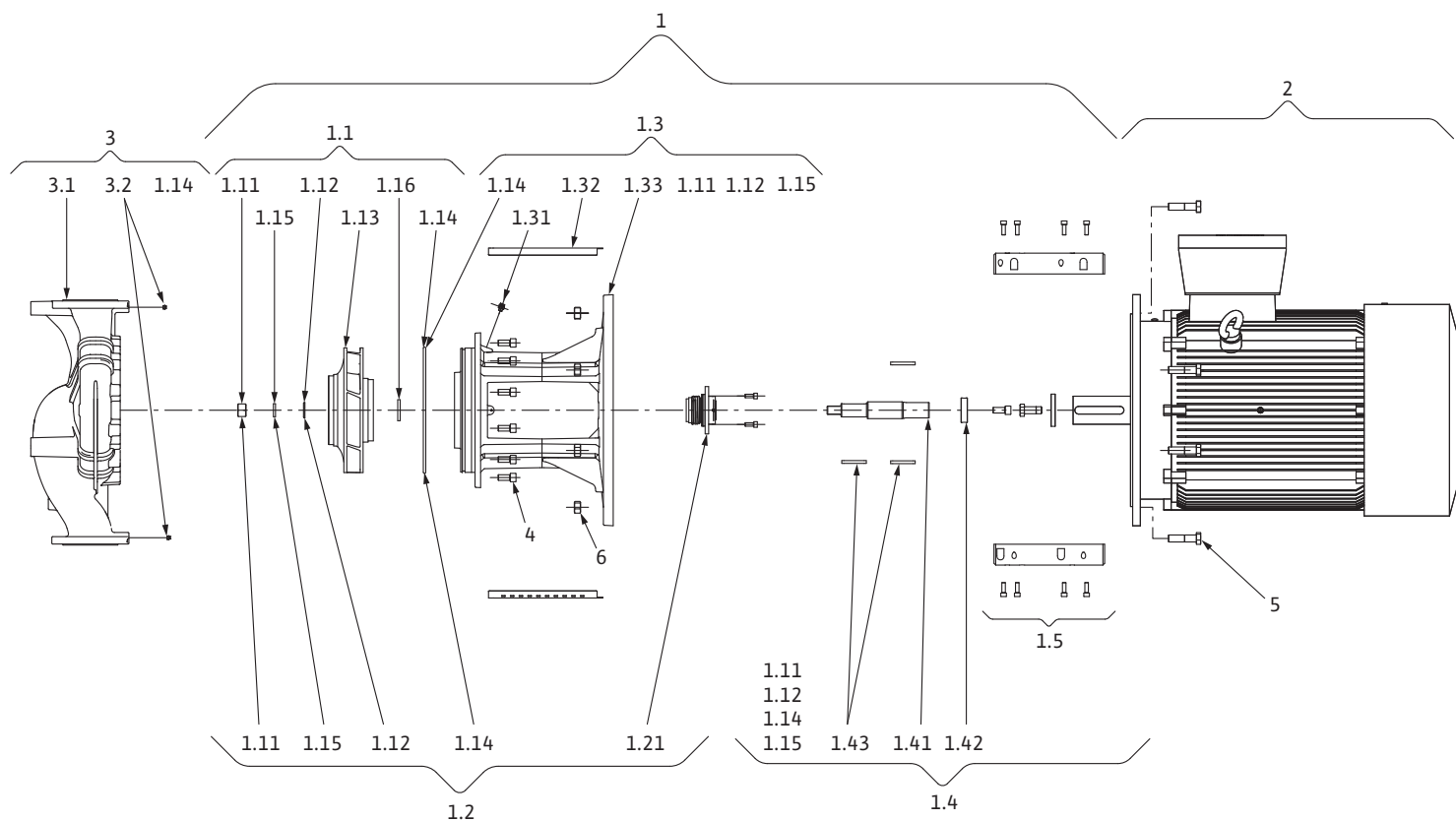


Fig. 30: BL Back Pull-Out

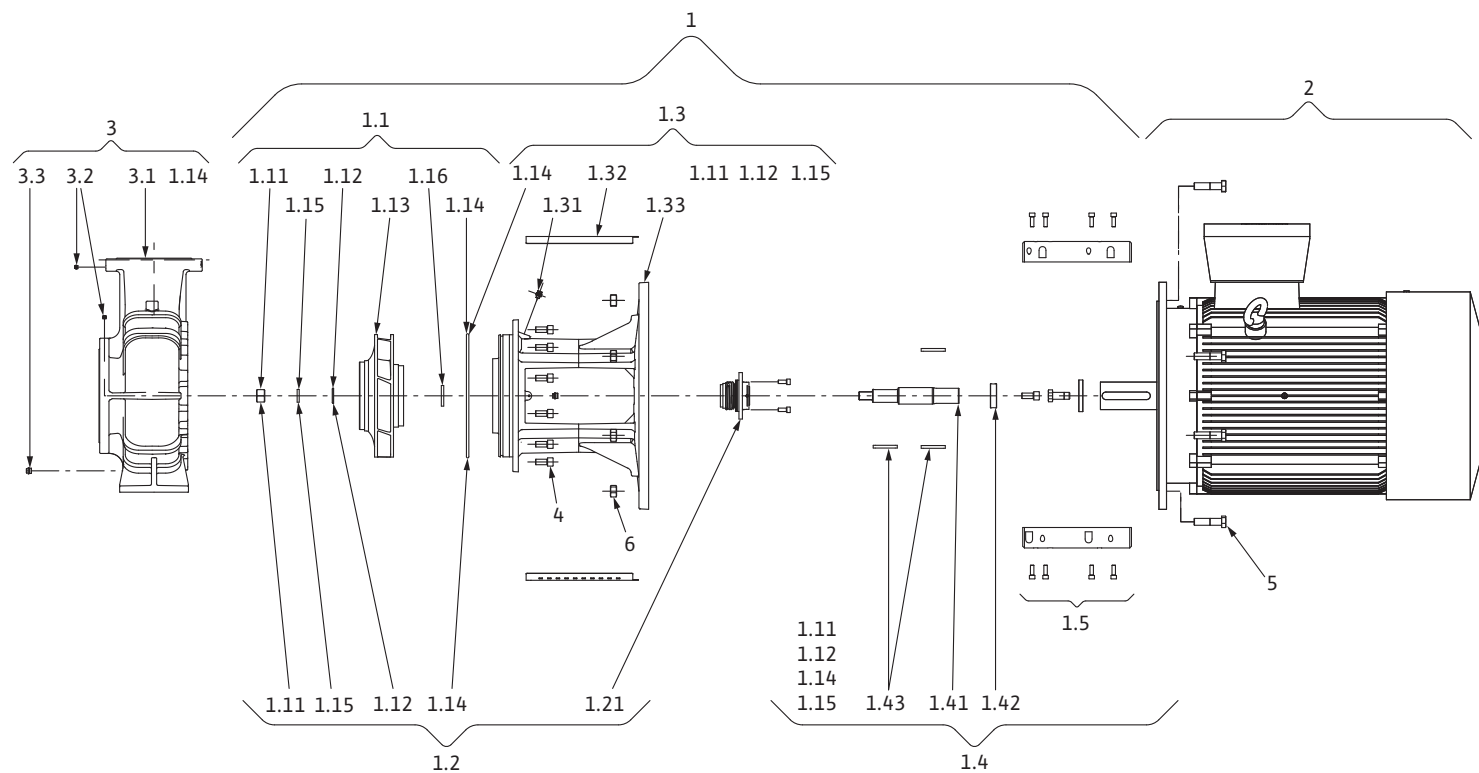
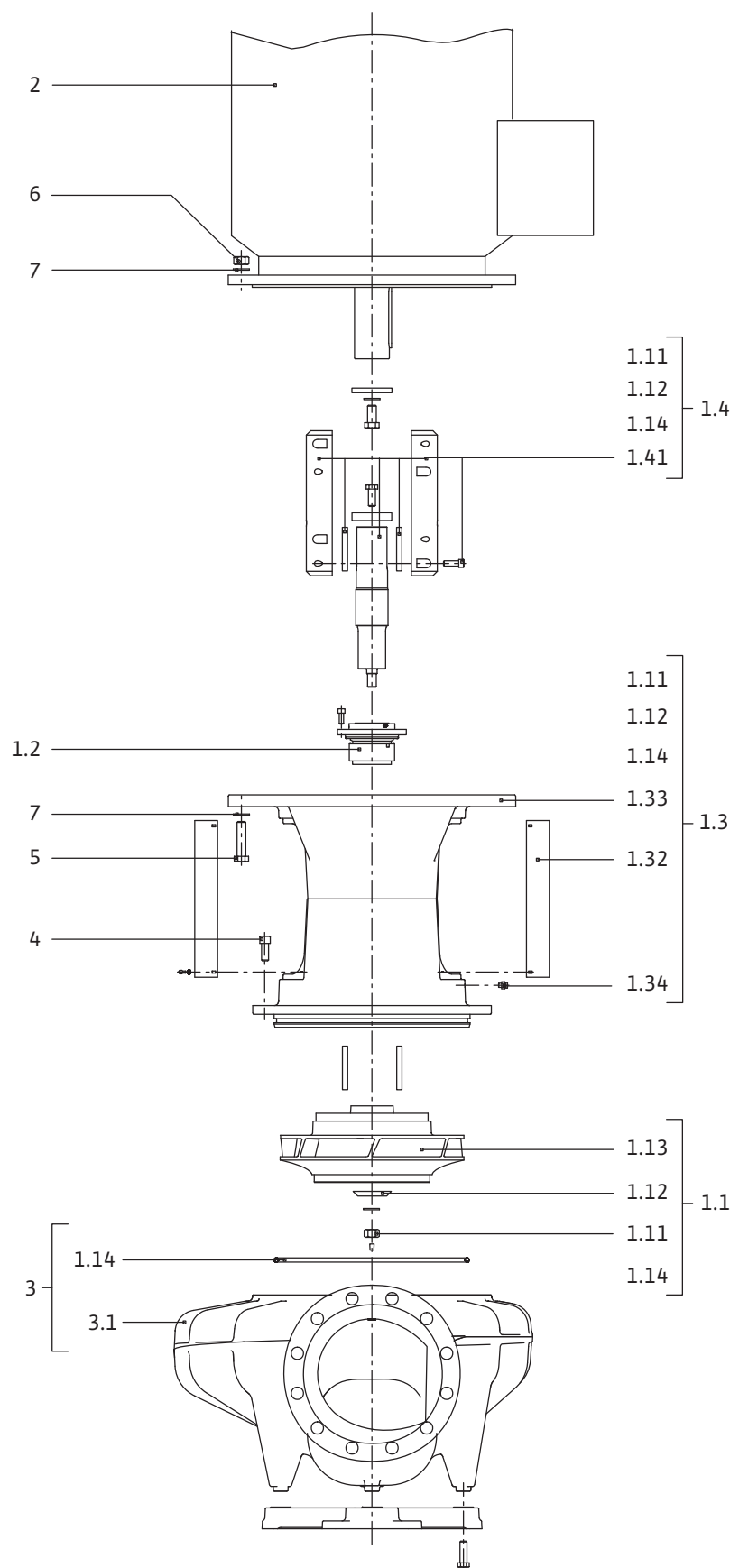


Fig. 31: IL 250



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1 General information

About this document

The language of the original operating instructions is German. All other languages of these instructions are translations of the original operating instructions.

These installation and operating instructions are an integral part of the product. They must be kept readily available at the place where the product is installed. Strict adherence to these instructions is a precondition for the proper use and correct operation of the product.

These installation and operating instructions correspond to the relevant version of the product and the underlying safety standards valid at the time of going to print.

EC declaration of conformity:

A copy of the EC declaration of conformity is a component of these operating instructions.

If a technical modification is made on the designs named there without our agreement or the declarations made in the installation and operating instructions on product/personnel safety are not observed, this declaration loses its validity.

2 Safety

These operating instructions contain basic information which must be adhered to during installation, operation and maintenance. For this reason, these operating instructions must, without fail, be read by the service technician and the responsible specialist/operator before installation and commissioning.

It is not only the general safety instructions listed under the main point "safety" that must be adhered to but also the special safety instructions with danger symbols included under the following main points.

2.1 Symbols and signal words in the operating instructions

Symbols



General danger symbol



Danger due to electrical voltage



NOTE

Signal words

DANGER!

Acutely dangerous situation.

Non-observance results in death or the most serious of injuries.

WARNING!

The user can suffer (serious) injuries. 'Warning' implies that (serious) injury to persons is probable if this information is disregarded.

CAUTION!

There is a risk of damaging the product/unit. 'Caution' implies that damage to the product is likely if this information is disregarded.

NOTE:

Useful information on handling the product. It draws attention to possible problems.

	Information that appears directly on the product, such as • Direction of rotation arrow, • Connection markings • Rating plate, • Warning sticker, must be strictly complied with and kept in legible condition.
2.2 Personnel qualifications	The installation, operating and maintenance personnel must have the appropriate qualifications for this work. Area of responsibility, terms of reference and monitoring of the personnel are to be ensured by the operator. If the personnel are not in possession of the necessary knowledge, they are to be trained and instructed. This can be accomplished, if necessary, by the manufacturer of the product at the request of the operator.
2.3 Danger in the event of non-observance of the safety instructions	Non-observance of the safety instructions can result in risk of injury to persons and damage to the environment and the product/unit. Non-observance of the safety instructions results in the loss of any claims to damages. In detail, non-observance can, for example, result in the following risks: • Danger to persons due to electrical, mechanical and bacteriological factors, • Damage to the environment due to leakage of hazardous materials, • Property damage, • Failure of important product/unit functions, • Failure of required maintenance and repair procedures.
2.4 Safety consciousness on the job	The safety instructions included in these installation and operating instructions, the existing national regulations for accident prevention together with any internal working, operating and safety regulations of the operator are to be complied with.
2.5 Safety instructions for the operator	This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety. • Children should be supervised to ensure that they do not play with the appliance. • If hot or cold components on the product/unit lead to hazards, local measures must be taken to guard them against touching. • Guards protecting against touching moving components (such as the coupling) must not be removed whilst the product is in operation. • Leakages (e.g. from a shaft seal) of hazardous fluids (e.g. explosive, toxic or hot) must be conveyed away so that no danger to persons or to the environment arises. National statutory provisions are to be complied with. • Highly flammable materials are always to be kept at a safe distance from the product. • Danger from electrical current must be eliminated. Local directives or general directives [e.g. IEC, VDE etc.] and instructions from local energy supply companies must be adhered to.

2.6 Safety instructions for installation and maintenance work

The operator must ensure that all installation and maintenance work is carried out by authorised and qualified personnel, who are sufficiently informed from their own detailed study of the operating instructions.

Work on the product/unit must only be carried out when at a standstill. It is mandatory that the procedure described in the installation and operating instructions for shutting down the product/unit be complied with.

Immediately on conclusion of the work, all safety and protective devices must be put back in position and/or recommissioned.

2.7 Unauthorised modification and manufacture of spare parts

Unauthorised modification and manufacture of spare parts will impair the safety of the product/personnel and will make void the manufacturer's declarations regarding safety.

Modifications to the product are only permissible after consultation with the manufacturer. Original spare parts and accessories authorised by the manufacturer ensure safety. The use of other parts will absolve us of liability for consequential events.

2.8 Improper use

The operational reliability of the supplied product is only guaranteed for conventional use in accordance with chapter 4 of the installation and operating instructions. The limit values must on no account fall under or exceed those values specified in the catalogue/data sheet.

3 Transport and interim storage

3.1 Shipping

The pump is delivered from the factory packaged in a cardboard box or secured to a pallet and protected against dust and moisture.

Transport inspection

On arrival, inspect the pump immediately for any transport damage. If transport damage is detected, the necessary steps involving the carrier must be taken within the specified period.

Storage

Before installation, the pump must be kept dry, frost-free and protected from mechanical damage.

If available, leave the covers on the pipe connections so that no dirt and other foreign matter can get into the pump housing.

Rotate the pump shaft once a week to prevent scoring at the bearings and sticking.

Consult Wilo about which corrosion-proofing measures are to be carried out in the event of a long storage time.



CAUTION! Risk of damage due to incorrect packaging!

If the pump is transported again at a later time, it must be packaged so that it cannot be damaged during transport.

- Use the original packaging for this, or select equivalent packaging.

3.2 Transport for installation/dismantling purposes



WARNING! Risk of injury!

Improper transport can lead to personal injury.

- Unload boxes, lathed spaces, pallets or cartons, depending on their size and construction, with forklift trucks or with slings.
- Always lift heavy parts of more than 30 kg with hoisting gear that is in accordance with local regulations. The bearing capacity has to be adapted to the weight.

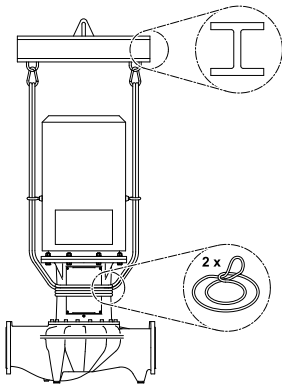


Fig. 32: Fitting load slings
(Version IL)

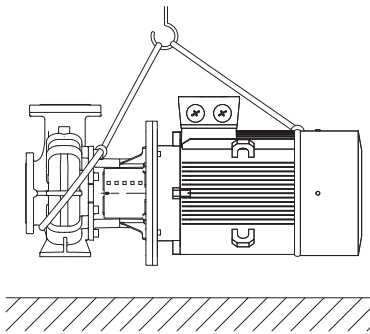


Fig. 33: Fitting load slings
(Version BL)

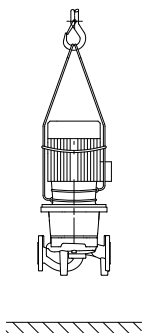


Fig. 34: Transporting the pump

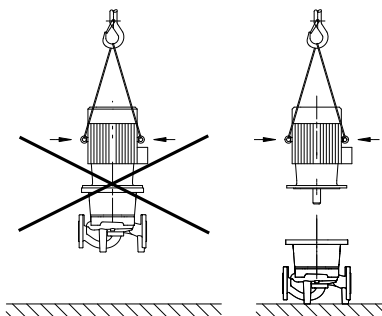


Fig. 35: Transporting the motor

- The pump must be transported using approved lifting gear (e.g. block and tackle, crane, etc.). This must be secured to the pump flanges and, if necessary, to the outer diameter of the motor (protection against slipping is required!).
- When lifting machines or parts using eyes, only use load hooks or shackles that are in accordance with local regulations.
- To lift with the crane, the pump must be supported by suitable belts as shown in (Fig. 32/33). Place loops around the pump which tighten from the pump's own weight.
- The transport eyes on the motor are only for guiding while bearing the load (Fig. 34).
- The transport eyes on the motor are only for transporting the motor and are not approved for transporting the complete pump (Fig. 35).
- If load chains or ropes are put over sharp edges, a guard has to be used or they have to be put through eyes.
- When using a pulley block and tackle or other hoisting gear, be sure that the load is lifted vertically.
- Prevent the suspended load from swinging. This can be done, for example, by using a second pulley block and tackle, whereby the pulling direction of the two pulley blocks should be less than 30° to vertical.
- Never subject load hooks, eyes or shackles to bending forces – their load axes have to be in the direction of the tractive forces!
- When lifting, be sure that the load limit of a rope is reduced for diagonal pulling. The safety and effectiveness of a stranding is best ensured when all load-bearing elements are loaded in the vertical direction to the greatest extent possible. If need be, use a lifting arm which can be attached vertically to the load rope.
- Set up a safety zone in such a way that there is no danger in the event that the load or a part of the load slips or the hoisting gear breaks or tears.
- Never leave a load longer than necessary in a suspended position! Be sure there is no danger to personnel when accelerating and slowing down during the lifting procedure.



WARNING! Risk of injury!

Setting up the pump without securing it can lead to personal injury.

- Do not place the pump unsecured on the pump support feet. The base with the threaded holes is only used for attachment. When standing freely, the pump might not be sufficiently stable.



DANGER! Risk of fatal injury!

The pump itself and pump parts can be extremely heavy. Falling parts pose a risk of cuts, crush injuries, bruises or impacts, which may lead to death.

- Always use suitable lifting equipment and secure parts against falling.
- Never stand underneath a suspended load.
- Make sure the pump is securely positioned and is stable during storage and transport as well as prior to all installation and other assembly work.
- Always wear protective clothing, protective gloves and protective goggles when working.

4 Intended use

Purpose

The glanded pumps of the IL (in-line single pump) and BL (monobloc pumps) series are meant to be used as circulators in building services.

Fields of application

They may be used for:

- Hot-water heating systems
- Cooling and cold water circulation systems
- Process water systems
- Industrial circulation systems
- Heat carrier circuits

Contraindications

Typical installation locations are technical rooms within the building with other domestic installations. No provision has been made for direct installation of the device in rooms used for other purposes (residential and work rooms).

For these series, it is possible to set them up outdoors only in their corresponding, special version, on request (see chapter 7.3 "Anti-condensation heater connection" on page 46).



CAUTION! Risk of property damage!

Non-permitted substances in the fluid can destroy the pump.

Abrasive solids (e.g. sand) increase pump wear.

Pumps without an Ex rating are not suitable for use in potentially explosive areas.

- Intended use includes compliance with this manual.
- Any other use is regarded as non-compliant with the intended use.

5 Product information

5.1 Type key

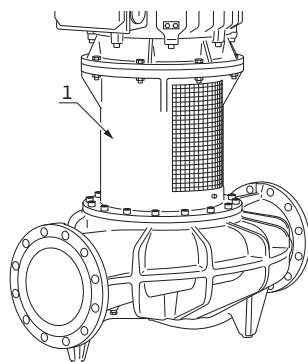


Fig. 36: Arrangement of the pump rating plate

The type key consists of the following elements:

Example: IL 250/420-110/4 BL 125/315-45/4	
IL	Flange-end pump as In-line single pump
BL	Flange-end pump as Bloc pump
250	Nominal diameter DN of the pipe connection (for BL: pressure side) [mm]
420	Nominal impeller diameter [mm]
110	Rated power P_2 [kW]
4	Number of poles, motor

Pump type rating plate:

Fig. 36, item 1 shows the arrangement of the pump rating plate.

5.2 Technical data

Property	Value	Remarks
Rated speed	50 Hz version • IL/BL (2-/4-pole): 2900/1450 rpm	Depending on pump type
	60 Hz version • IL/BL (2-/4-pole): 3480/1750 rpm	Depending on pump type
Nominal diameters DN	IL: 32 to 200 mm BL: 32 to 150 mm (pressure side)	
Pipe and pressure measurement connections	PN 16 flange in accordance with DIN EN 1092-2 with pressure measurement connections Rp 1/8 in accordance with DIN 3858. Partially flange PN 25, depending on the pump type	
Permissible min./max. fluid temperature	-20 °C to +140 °C	Depending on fluid
Min./max permissible ambient temperature	0 to 40 °C	Lower or higher ambient temperatures on request
Storage temperature min./max.	-20 °C to +60 °C	
Maximum permitted operating pressure	16 bar (version...-P4: 25 bar)	Version...-P4 (25 bar) as special version at additional charge (availability depends on the pump type)
Insulation class	F	
Protection class	IP55	
Approved fluids	Heating water according to VDI 2035 Process water Cooling/cold water Water-glycol mixtures up to 40 vol.-%	Standard version Standard version Standard version Standard version
	Heat transfer oil	Special version or auxiliary equipment (at additional charge)
	Other media upon request	Special version or auxiliary equipment (at additional charge)
Electrical connection	3~ 400 V, 50 Hz	Standard version
	3~ 230 V, 50 Hz (up to 3 kW inclusive)	Alternative application for standard version (no additional charge)
	3~ 230 V, 50 Hz (from 4 kW)	Special version or auxiliary equipment (at additional charge)
	3~ 380 V, 60 Hz	partially standard version
Special voltage/frequency	Pumps with motors with different voltages or with other frequencies are available on request	Special version or auxiliary equipment (at additional charge)
PTC thermistor sensor	standard version from 75 kW	
Speed control, pole switchover	Wilo control devices (e.g. Wilo-CC/SC-HVAC-System)	Standard version
	Pole switchover	Special version or auxiliary equipment (at additional charge)

Table 1: Technical data

Fluids

If water–glycol mixtures are used (or fluids with a viscosity other than that of pure water), an increase in power consumption of the pump is to be taken into account. Only use mixtures with corrosion protection inhibitors. The respective manufacturer's instructions are to be observed.

- Adjust the motor power if required!
- The fluid must be sediment-free.
- Wilo's approval must be obtained for the use of other fluids.
- In systems built according to the state of the art, it can be assumed under normal system conditions that the standard seal/standard mechanical seal is compatible with the fluid. Special circumstances (e.g. solid material, oils or EPDM-corrosive substances in the fluid, air in the system etc.) may require special seals.



NOTE:

Always read and follow the material safety data sheet for the fluid being pumped!

5.3 Scope of delivery

- Pump IL/BL
(IL 250 including mounting foot for installation and foundation fixation)
- Installation and operating instructions

5.4 Accessories

Accessories must be ordered separately:

- PTC thermistor tripping unit for switch cabinet installation
- BL: For installation on a base or baseplate for rated power of 5.5 kW and above

For a detailed list, consult the catalogue and spare parts documentation.

6 Description and function

6.1 Description of the product

All the pumps described are single-stage low-pressure centrifugal pumps in compact construction design with a coupled motor. The mechanical seal is maintenance-free. The pumps can be installed both directly as a pipe installation pump in a sufficiently anchored pipe or placed on a foundation base. Installation options depend on the size of the pump.

In conjunction with a control device (e.g. Wilo-CC/SC-HVAC-System), performance of the pumps can also be continuously controlled. This allows optimisation of the pump output for the demands of the installation and economically efficient pump operation.

Version IL:

The pump housing has an in-line design, i.e. the flanges on the suction and pressure sides lie along a centre line (Fig. 37). All pump housings are provided with pump support feet. Installation on a foundation base is recommended for rated powers of 5.5 kW and higher.

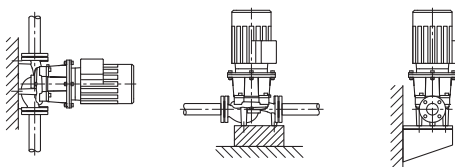


Fig. 37: View of IL

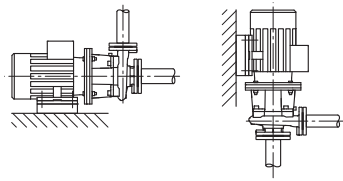


Fig. 38: View of BL

6.2 Anticipated noise levels

Version BL:

Spiral housing pump with flange dimensions in accordance with DIN EN 733 (Fig. 38). Depending on the construction:

Up to 4 kW motor power: Pedestal screwed onto the pump or cast feet on the pump housing.

From 5.5 kW motor power: Motors with cast or screwed on feet. Version in design B: With cast feet on the pump housing.

Motor power P_N [kW]	Sound-pressure level L_p (A) [dB(A)] ¹⁾	
	2900 rpm IL, BL	1450 rpm IL, BL
37	77	70
45	72	72
55	77	74
75	77	74
90	77	72
110	79	72
132	79	72
160	79	74
200	79	77
250	85	–

¹⁾ Spatial mean value of sound-pressure levels on a square measuring surface at a distance of 1 m from the surface of the motor.

Table 2: Anticipated noise levels

6.3 Permissible forces and torques on the pump flanges (only BL pumps)

See Fig. 39 and list "Table 3: Permissible forces and torques on the pump flanges" on page 39.

Values in acc. with ISO/DIN 5199 – class II (2002) – Appendix B, Family no. 1A.

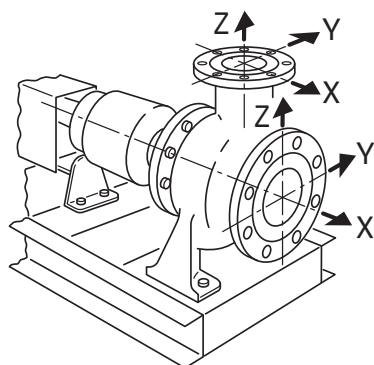


Fig. 39: Permissible forces and torques on the pump flanges – pump made of grey cast iron

	DN	Forces F [N]				Torques M [Nm]			
		F _X	F _Y	F _Z	Σ Forces F	M _X	M _Y	M _Z	Σ Torques M
Pressure port	32	315	298	368	578	385	263	298	560
	40	385	350	438	683	455	315	368	665
	50	525	473	578	910	490	350	403	718
	65	648	595	735	1155	525	385	420	770
	80	788	718	875	1383	560	403	455	823
	100	1050	945	1173	1838	613	438	508	910
	125	1243	1120	1383	2170	735	525	665	1068
Suction port	150	1575	1418	1750	2748	875	613	718	1278
	50	578	525	473	910	490	350	403	718
	65	735	648	595	1155	525	385	420	770
	80	875	788	718	1383	560	403	455	823
	100	1173	1050	945	1838	613	438	508	910
	125	1383	1243	1120	2170	735	525	665	1068
	150	1750	1575	1418	2748	875	613	718	1278
	200	2345	2100	1890	3658	1138	805	928	1680

Table 3: Permissible forces and torques on the pump flanges

If not all working loads reach the maximum permitted values, one of these loads may exceed the normal limit value under the condition that following additional conditions are fulfilled:

- All force and torque components must be limited to 1.4 times of the maximum permitted value.
- The following equation applies to the actual forces and torques acting on every flange (the following condition must be fulfilled):

$$\left(\frac{\sum |F|_{\text{actual}}}{\sum |F|_{\text{max.permitted}}} \right)^2 + \left(\frac{\sum |M|_{\text{actual}}}{\sum |M|_{\text{max.permitted}}} \right)^2 \leq 2$$

Where the total load $\sum |F|$ and $\sum |M|$ is the arithmetic total of every flange (inlet and outlet), for the actual as well as the maximum permitted values, without taking into consideration their algebraic sign, at the pump level (inlet flange + outlet flange).

7 Installation and electrical connection

Safety



DANGER! Risk of fatal injury!

Incorrect installation and improper electrical connections can be life-threatening.

- Electrical connections should only be established by qualified electricians and in compliance with the applicable regulations!
- Adhere to accident prevention regulations!



DANGER! Risk of fatal injury!

Failure to install safety devices on the terminal box or near the coupling can cause electrical shock or contact with rotating parts, potentially resulting in life-threatening injuries.

- Before commissioning, all safety devices (such as terminal box covers or coupling covers) that were removed must be reinstalled.



DANGER! Risk of fatal injury!

The pump itself and pump parts can be extremely heavy. Falling parts pose a risk of cuts, crush injuries, bruises or impacts, which may lead to death.

- Always use suitable lifting equipment and secure parts against falling.
- Never stand underneath a suspended load.
- Make sure the pump is securely positioned and is stable during storage and transport as well as prior to all installation and other assembly work.



CAUTION! Risk of property damage!

Risk of damage due to incorrect handling.

- Have the pump installed by qualified personnel only.



CAUTION! Damage to the pump due to overheating!

The pump must not be allowed to operate dry for more than 1 minute. Dry running causes a build-up of energy in the pump, which can damage the shaft, impeller, and mechanical seal.

- Make sure that the volume flow Q_{min} does not fall below the minimum value.

Calculation of Q_{min} :

$$Q_{min} = 10 \% \times Q_{max \text{ pump}}$$

7.1 Installation

Preparation

- The pump must be checked for compliance with the specifications on the delivery receipt; any damages or missing parts must be conveyed to Wilo immediately. Check slat crates/boxes/wrappings for spare parts or accessory components which could be included with the pump.

Installation location

- The pumps must be protected from the weather and installed in a frost/dust-free, well-ventilated and vibration-insulated environment which is not potentially explosive. The pump must not be installed outdoors.
- Install the pump in a place that is easy to access so that later inspections, maintenance (e.g. the mechanical seal) or replacement is easily possible.
- Provide space for minimum axial distance between a wall and the fan cover of the motor: Free removal dimension of at least 200 mm + diameter of the fan cover.

Base

- For a vibration-insulated installation, a simultaneous isolation of the base block itself from the building structure by an elastic intermediate layer (e.g. cork or Mafund pad) is required for few pump types.



CAUTION! Risk of property damage!
Danger of damage due to incorrect handling.

- A missing base or an incorrect installation of the unit on the base can lead to a malfunction of the pump; this is excluded from the guarantee.

Positioning/alignment

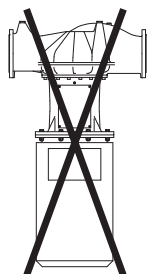
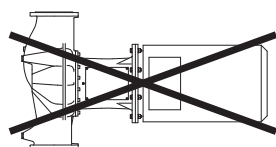
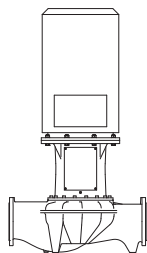


Fig. 40: Version IL:
Permitted/impermissible installation
positions

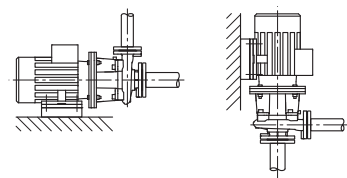


Fig. 41: Version BL



CAUTION! Risk of property damage!
Risk of damage due to incorrect handling.

- Only use lifting eyes on the motor for carrying the weight of the motor and not for carrying the entire pump.
- Lift the pump using permitted lifting gear (see chapter 3 "Transport and interim storage" on page 33.)



NOTE:

Shut-off devices shall be installed in front of and behind the pump in all cases, in order to avoid having to drain the entire system when checking, servicing or replacing the pump. If need be, the necessary non-return valves are to be provided.

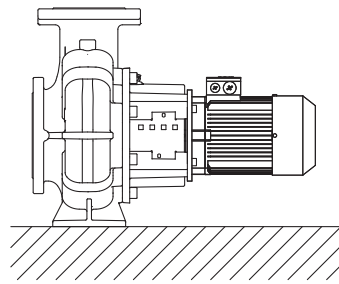
- The lantern has an opening on the underside which allows a drainage pipe for condensation water/condensate to be connected (e.g. when used in air-conditioning or cooling systems). The condensate generated can then be effectively drained.
- The pipes and pump must be free of mechanical stress when installed. The pipes must be fixed in such a way that the pump is not supporting the weight of the pipes.
- The air vent valve (Fig. 29 /30/31, item 1.31) must always face upwards.
- Installation position: Only vertical installation is permitted (see Fig. 40).
- BL series monobloc pumps are to be mounted on a sufficiently strong base or mounting brackets (Fig. 41). For pumps of BL type, the motor must be supported from a motor power of 18.5 kW, see installation examples BL (Fig. 42).

Only pump type design B: For motor power from 37 kW four pole or 45 kW two pole, the pump housing and motor must be supported. Suitable supports from the Wilo accessories range can be used for this purpose.

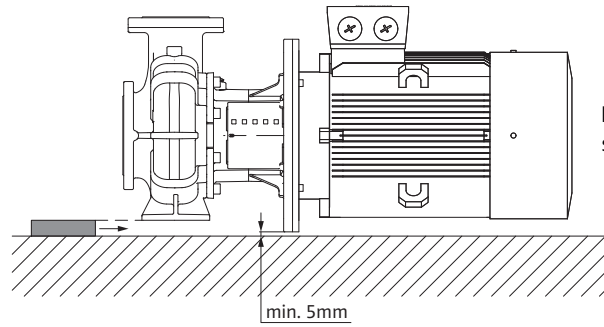


NOTE:

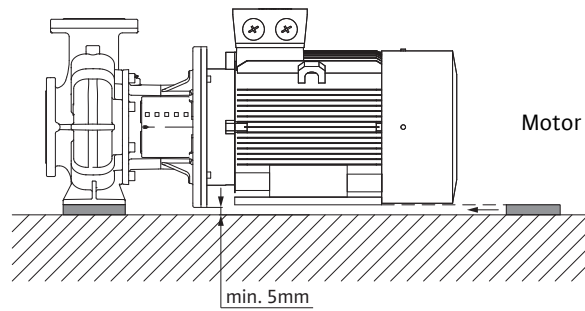
The motor terminal box must not face downward. If necessary, the motor or motor impeller unit can be rotated once the screws have been loosened. While rotating the motor or motor impeller unit, ensure that the housing O-ring seal does not become damaged.



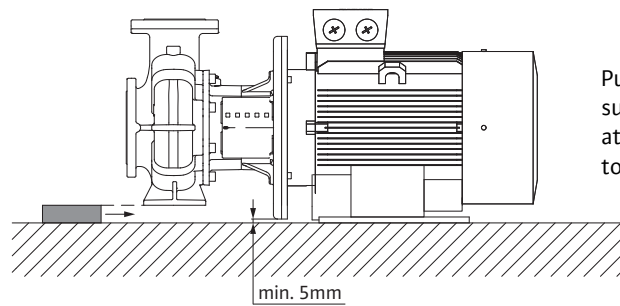
No
support
required



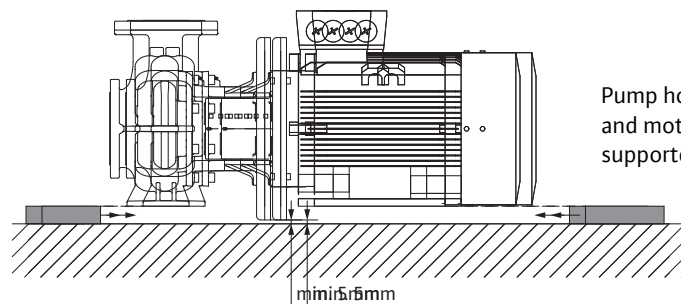
Pump housing
supported



Motor supported



Pump housing
supported, motor
attached
to a foundation



Pump housing
and motor
supported

Fig. 42: Installation examples BL



CAUTION! Risk of property damage!
Risk of damage due to incorrect handling.

- When pumping out a tank, ensure that the fluid level is always high enough above the suction port of the pump so that the pump never runs dry. The minimum inlet pressure must be observed.



NOTE:

In the case of insulated systems, only the pump housing may be insulated, not the lantern or drive.

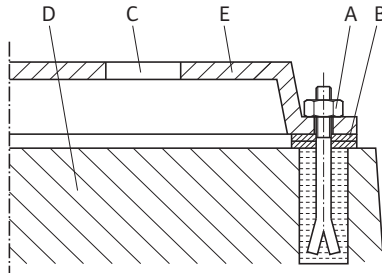


Fig. 43: Example of a foundation screwed connection

Example of a foundation screwed connection (Fig. 43):

- Align the whole unit when installing on the foundation with the help of the spirit level (at the shaft/pressure port).
- Always put the washers (B) to the left and right directly at the fixation material (e.g. stone bolts (A)) between the baseplate (E) and foundation (D).
- Evenly and firmly tighten the fixation material.
- For distances > 0.75 m, support the baseplate at the centre between the fastening elements

Connection of the pipes



CAUTION! Risk of property damage!
Risk of damage due to incorrect handling.

- Under no circumstances may the pump be used as a fixed point for the pipe.
- The pipes and pump must be free of mechanical stress when installed. The pipes must be fixed in such a way that the pump is not supporting the weight of the pipes.
- The present NPSH of the system always has to be greater than the required NPSH of the pump.
- The forces and torques being exerted by the pipe system on the pump flange (e.g., by warping, thermal expansion) may not exceed the permitted forces and torques.
- Brace the pipes directly before the pump and connect them without tension. Their weight must not put a strain on the pump.
- The inlet pipe is to be kept as short as possible. Lay the inlet pipe to the pump so that it continuously rises while the intake falls. Avoid possible air entry points.
- If a dirt trap in the inlet pipe is required, then its free cross-section has to be 3 – 4 times the cross-section of the piping.
- For short pipes, the nominal diameters should be at least that of the pump connections. The long pipes, the most economical nominal diameter is to be determined on a case-by-case basis.
- Adapters for larger nominal diameters should be made with a flaring angle of about 8° to prevent greater pressure losses.



NOTE:

Shut-off devices shall be installed in front of and behind the pump in all cases, in order to avoid having to drain the entire system when checking, servicing or replacing the pump. If required, a non-return valves is to be provided.

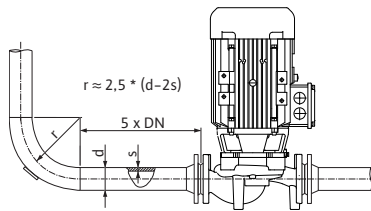


Fig. 44: Settling section before and after the pump



NOTE:

A settling section must be provided before and after the pump, in the form of a straight pipe. The length of this settling section should be at least 5 x DN of the pump flange (Fig. 44). This measure serves to avoid flow cavitation.

- Only connect the pipes after the conclusion of all welding and soldering work as well as the cleaning/flushing of the system. Dirt can cause the pump to fail.
- Remove the flange covers at the suction and pressure ports of the pump before attaching the piping.

Final check

Check the alignment of the unit again as described in chapter 7.1 "Installation" on page 40.

- Tighten the foundation bolts if necessary.
- Check all connections for correctness and function.
- It must be possible to turn the coupling/shaft by hand.

Do not allow the coupling/shaft to turn:

- Release the coupling and re-tighten.

In case this measure is not successful:

- Remove the motor (see chapter 9.2.3 "Changing the motor" on page 52).
- Clean the motor centring and flange.
- Reinstall the motor.

7.2 Electrical connection

Safety



DANGER! Risk of fatal injury!

Incorrect electrical connections can cause fatal electric shocks.

- **Have the electrical connection set up by an electrician approved by the local electricity supplier only and in accordance with local regulations.**
- **Observe the installation and operating instructions for the accessories!**



DANGER! Risk of fatal injury!

Contact voltage hazardous to human life.

Work on the terminal box may only be started once 5 minutes have passed, due to the dangerous residual contact voltage (capacitors).

- **Before working on the pump, disconnect the supply voltage and wait for 5 minutes.**
- **Check whether all connections (including potential-free contacts) are voltage-free.**
- **Never use an object to poke around the openings on the terminal box or motor and never insert anything into the module or motor!**



WARNING! Risk of mains overload!

An inadequate mains design can lead to system failures and cable fires due to mains overload.

- **When designing the mains, with regard to the used cable cross-sections and fuses, pay special attention to the fact that, in multi-pump operation, it is possible that all pumps operate at the same time for short periods.**

Preparation/notes

- The electrical connection must be made in accordance with VDE 0730/part 1 via a fixed mains connection cable, which is provided with a plug device or an all-pole switch with a contact opening width of at least 3 mm.
- In order to ensure drip protection and strain relief on the threaded cable connection, cables are to be used which have a sufficient outer diameter and are to be screwed sufficiently tightly.

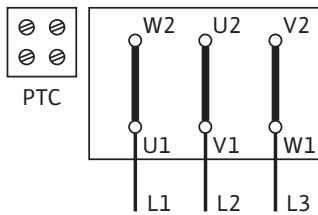


Fig. 45: Y-Δ-start (standard)

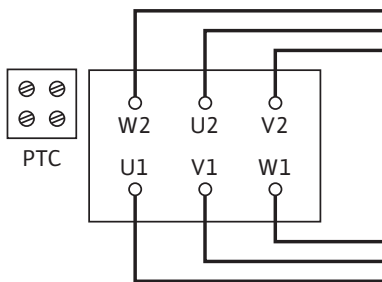


Fig. 46: Δ connection

- To get rid of any drips that accumulate, the cables are to be bent into a drain loop near the threaded cable connection.
- By positioning the threaded cable connection or by laying the cables accordingly, ensure that no drips can enter the terminal box. Non-assigned threaded cable connections must remain sealed with the plugs provided by the manufacturer.
- The connection pipe is to be placed in such a way that it can under no circumstances come into contact with the pipe and/or the pump and motor housing.
- When pumps are used in systems with water temperatures above 90 °C, a suitably heat-resistant connection pipe must be used.
- Check the current type and voltage of the mains connection.
- Observe the rating plate information for the pump. The current type and voltage of the mains connection must correspond to the details on the rating plate.
- Fuse protection on mains side: Dependent on rated motor current.
- Take additional earthing into account!
- The motor must be secured against overloading using a motor protection switch or the PTC thermistor tripping unit (see chapter 5.4 "Accessories" on page 37).

**NOTE:**

The connection diagram for the electrical connection is in the terminal box (also see Fig. 45/46).

Setting the motor protection switch:

- Adjust according to the rated motor current specified on the name plate of the motor.
- Y-Δ start: If the motor protection switch is switched in the supply line to a Y-Δ contactor combination, adjust the switch as for direct starting.
- If the motor protection switch is switched in a thread of the motor supply line (U1/V1/W1 or U2/V2/W2), set the motor protection switch to 0.58 x rated motor current.
- The special version of the motor is equipped with PTC thermistor sensors. Connect the PTC thermistor sensors to the PTC thermistor tripping unit.

**CAUTION! Risk of property damage!****Risk of damage due to incorrect handling.**

- **Only a maximum voltage of 7.5 V DC may be applied to the PTC thermistor sensor terminals. A higher voltage will destroy the PTC thermistor sensor.**
- The mains connection to the terminal board is dependent on the motor power P_2 , the mains voltage and the activation type. The recommended switching arrangement of the connection bridges in the terminal box should be taken from the following list "Table 4: Connection terminal assignment" on page 46 and Fig. 45/46.
- When connecting automatic switchgears, observe the relevant installation and operating instructions.
- For three-phase current motors with Y-Δ connection, be sure that the switch-over points between star and triangle are very close together in time. Longer switch-over times can lead to pump damage.

Required connection of the connecting bridges in the terminal box:

Activation type	Mains voltage 3~ 400 V
Y-Δ start (default)	Remove connection bridges (Fig. 45)
Start via Soft starter	Δ connection (Fig. 46)

Table 4: Connection terminal assignment

- When connecting automatic switchgears, observe the relevant installation and operating instructions.
- For three-phase current motors with Y-Δ connection, be sure that the switch-over points between star and triangle are very close together in time. Longer switch-over times can lead to pump damage.

Recommendation for setting the times for Y-Δ activation:

Motor power	Y time to be set
> 30 kW	< 5 s



CAUTION! Risk of property damage!
Risk of damage due to incorrect handling.

- **Only carry out the direction of rotation monitoring after the system is filled. Even brief dry running will destroy the mechanical seal.**



NOTE:

Use of soft starters is recommended to limit the starting current and preventing tripping of excess current protection devices.

7.3 Anti-condensation heater connection

We recommend an anti-condensation heater for motors which, due to climatic conditions, are at a risk of condensation (e.g. stationary motors in moist atmospheres, or motors exposed to extreme variations in temperature). Corresponding motor variations, which are equipped with an anti-condensation heater at the factory, can be ordered as a special version. The anti-condensation heater is used to protect the motor windings against condensation water inside the motor.

- The anti-condensation heater is connected to the terminals HE/HE in the terminal box (connection voltage: 1~ 230 V/50 Hz).



CAUTION! Risk of property damage!
Risk of damage due to incorrect handling.

- **The anti-condensation heater may not be switched on during motor operation.**

8 Commissioning

Safety



DANGER! Risk of fatal injury!

Failure to install safety devices on the terminal box or near the coupling can cause electrical shock or contact with rotating parts, potentially resulting in life-threatening injuries.

- Before commissioning, all safety devices (such as terminal box covers or coupling covers) that were removed must be reinstalled.
- Keep a safe distance during commissioning!



WARNING! Danger of injury!

If the pump/system is installed improperly, fluid may be ejected during commissioning. Individual components may also become loose.

- Keep a safe distance from the pump during commissioning.

- **Wear protective clothing, protective gloves and protective goggles.**



NOTE:

It is recommended to have the pump serviced and checked by Wilo customer service.

Preparation

8.1 Initial commissioning

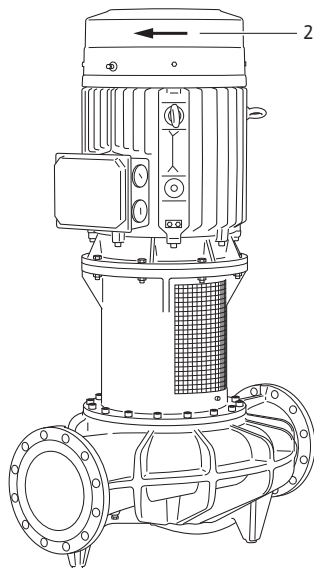


Fig. 47: Check direction of rotation

The pump has to reach ambient temperature before commissioning.

- Check whether the shaft can be rotated without any grinding. In case the impeller is blocked or grinds, loosen the coupling screws and tighten them with the specified torque (see list "Table 5: Screw tightening torques" on page 54).
- Prime and vent the unit correctly.



WARNING! Danger due to extremely hot or extremely cold pressurised fluid!

Depending on the temperature of the fluid and the system pressure, when the venting screw is opened completely, extremely hot or extremely cold fluid in liquid or vapour may escape or shoot out at high pressure.

- **Always exercise caution when opening the venting screw.**



CAUTION! Risk of property damage!

Running dry will destroy the mechanical seal.

- **Make sure that the pump does not run dry.**
- To avoid cavitation noises and damage, a minimum inlet pressure must be guaranteed at the suction port of the pump. This minimum inlet pressure depends on the operation situation and the duty point of the pump, and must be defined accordingly.
The main parameters for defining the minimum inlet pressure are the NPSH of the pump at its duty point and the vapour pressure of the fluid.
- By briefly switching on, check whether the direction of rotation agrees with the arrow on the fan cover (see Fig. 47, item 2). If the direction of rotation is incorrect, proceed as follows:
 - For direct starting: Swap 2 phases on the motor terminal board (e.g. L1 for phase L2).
 - For Y-Δ start, swap the winding start and winding end of 2 windings on the motor terminal board (e.g. V1 for V2 and W1 for W2).

8.1.1 Activation

- The unit may only be switched on when the shut-off device on the pressure side is closed! Only after full speed has been reached may this be slowly opened and be adjusted to the duty point.
- The unit must run smoothly and free of vibration.
- The mechanical seal ensures that a seal will not leak and requires no special setting. Should there be a small leakage at the beginning, it will stop when the initial inlet phase of the gasket is over.
- Immediately after the conclusion of all work, all the provided safety and protection equipment items must be properly installed and put into operation.



DANGER! Risk of fatal injury!

Failure to install safety devices on the terminal box or near the coupling can cause electrical shock or contact with rotating parts, potentially resulting in life-threatening injuries.

8.1.2 Switching off

- **Immediately after maintenance, all previously removed safety devices, such as terminal box covers or coupling covers, must be reinstalled.**



NOTE:

In case a non-return valve is installed in the pressure pipe, the shut-off device can remain open insofar as a counter pressure is present.



CAUTION! Risk of property damage!

Risk of damage due to incorrect handling.

- **When switching off the pump, the shut-off device in the inlet pipe may not be closed.**
- Switch off the motor and allow it to coast down completely. Ensure the coasting is smooth.
- For longer downtimes, close the shut-off device in the inlet pipe.
- For longer periods of non-use and/or danger of freezing, drain the pump and secure it against freezing.
- After removing the pump, keep it in a dry and dust-free storage.

8.2 Operation



NOTE:

The pump should always run smoothly and vibration-free and not be operated in conditions other than those specified in the catalogue/data sheet.



DANGER! Risk of burns or freezing to the pump when body parts come into contact with the pump!

Depending on the pump or system operating conditions (fluid temperature), the entire pump can become very hot or very cold.

- **Keep a safe distance during operation!**
- **In the case of high water temperatures and system pressures, allow the pump to cool down before all work.**
- **Always wear protective clothing, protective gloves and protective goggles when working.**



DANGER! Risk of fatal injury!

Due to protective equipment in the coupling area which has not been installed, contact with rotating parts can lead to deadly injuries.

- **Immediately after maintenance, all previously removed safety devices, such as terminal box covers or coupling covers, must be reinstalled.**
- **The pump may only be operated with the coupling guard metal sheets installed (Fig. 48, item 3).**
- Depending on the different operating conditions and the degree of installation's automation, the pump can be switched on and off in different ways. Observe the following:
 - Stop procedure:
 - Prevent return flow to the pump.
 - Do not work too long with an insufficient volume flow.
 - Start procedure:
 - Be sure that the pump is completely filled up.
 - Do not work too long with an insufficient volume flow.
 - Larger pumps require a minimum flow rate to operate properly.
 - Operating against a closed slide valve can lead to overheating in the centrifugal chamber and to damage of the shaft seal.
 - Ensure a continual flow to the pump with a sufficiently large NPSH.
 - Prevent insufficient counter pressure leading to a motor overload.

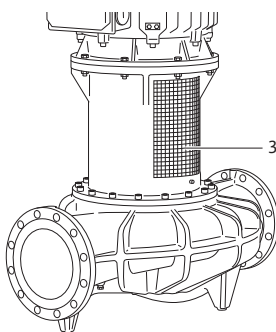


Fig. 48: Installed coupling guard metal sheet

- To prevent significant increases in motor temperature and excessive load on the pump, coupling, motor, gaskets and bearings, no more than 10 switch-on procedure should be performed per hour.

9 Maintenance

Safety

Have maintenance and repair work carried out by qualified personnel only!

It is recommended to have the pump serviced and checked by Wilo customer service.

When drafting a maintenance plan, expensive repairs can be avoided and a fault-free operation of the system can be achieved with a minimum of maintenance effort.



DANGER! Risk of fatal injury!

There is a risk of fatal injury from electric shock when working on electrical equipment.

- Work on electrical equipment may only be done by electricians approved by the local electricity supplier.
- Before working on electrical equipment, switch it off and secure it against being switched on again.
- Any damage to the pump connection cable should always be rectified by a qualified electrician only.
- Never use an object to poke around the openings on the terminal box or motor and never insert anything into the module or motor!
- Follow the installation and operating instructions for the pump, level control device and other accessories!



DANGER! Risk of fatal injury!

Failure to install safety devices on the terminal box or near the coupling can cause electrical shock or contact with rotating parts, potentially resulting in life-threatening injuries.

- Immediately after maintenance, all previously removed safety devices, such as terminal box covers or coupling covers, must be reinstalled.



DANGER! Risk of fatal injury!

The pump itself and pump parts can be extremely heavy. Falling parts pose a risk of cuts, crush injuries, bruises or impacts, which may lead to death.

- Always use suitable lifting equipment and secure parts against falling.
- Never stand underneath a suspended load.
- Make sure the pump is securely positioned and is stable during storage and transport as well as prior to all installation and other assembly work.



DANGER! Risk of burns or freezing to the pump when body parts come into contact with the pump!

Depending on the pump or system operating conditions (fluid temperature), the entire pump can become very hot or very cold.

- Keep a safe distance during operation!
- In the case of high water temperatures and system pressures, allow the pump to cool down before all work.
- Always wear protective clothing, protective gloves and protective goggles when working.

	 DANGER! Risk of fatal injury! The tools used during maintenance work on the motor shaft can be flung out if they come into contact with rotating parts, and cause serious or even fatal injuries. • The tools used during maintenance work must be removed completely before the pump is started up.
9.1 Air supply	The air supply to the motor housing must be checked at regular intervals. In case of contamination, ensure that an air supply is re-established in order to allow the motor to cool sufficiently.
9.2 Maintenance tasks	 DANGER! Danger of death! Falling pumps or pump parts may result in life-threatening injuries. • When performing installation work, protect the pump components against falling.  DANGER! Risk of fatal injury! There is a risk of fatal injury from electric shock when working on electrical equipment. • Check for absence of voltage and cover or cordon off adjacent live parts.
9.2.1 On-going maintenance	Replace all gaskets that have been removed for maintenance.
9.2.2 Replacing the mechanical seal	The mechanical seal is maintenance-free. There may be a slight amount of drip leakage during the running-in period. Even during normal operation of the pump, slight leakage involving occasional drops is common. Once in a while, however, a visual inspection is required. If there is clearly detectable leakage, the gasket is to be changed. Wilo offers a repair kit which contains the necessary parts for replacement.
Dismantling	Dismantling: • Disconnect the system from the power and secure it against being switched on. • Make sure it is not live. • Earth the work area and short-circuit. • Close the shut-off device upstream and downstream of the pump. • Disconnect the mains connection cable. • Depressurise the pump by opening the venting screw (Fig. 29 /30/31, item 1.31).  DANGER! Risk of scalding! Due to high fluid temperatures there is a danger of scalding. • At high fluid temperatures, let the pump cool down before starting any work.  NOTE: • When tightening screw connections in conjunction with the work described in the following: Observe the screw tightening torque for the thread type (see list "Table 5: Screw tightening torques" on page 54). • Dismantle the coupling guard metal sheets (Fig. 1, item 1). • Rotate the coupling/shaft in such a way that the four hexagon socket screws (twist guard; Fig. 2, item 1) are opposite to the cover/drilled holes. • Unscrew the hexagon socket screws (locking pins) one after the other far enough that the head is countersunk half way in the cover collar (Fig. 2 or Fig. 3, depending on the pump type). • Unscrew the 4 cover screws (Fig. 4).

- Screw in two of the cover screws up to the stop into the press-off drilled holes to press the cover out of its seat (Fig. 4/5).
- Unscrew one of the coupling screws and turn it all the way into one of the installation drilled holes (Fig. 6, item 1). To do this, the coupling half is fixed by the retaining washer (Fig. 6, item 3) at the impeller shaft.
- Unscrew the remaining coupling screws and take off the loose coupling half. If need be, use the provided press-off drilled holes (Fig. 6, item 4). The impeller shaft is now held up by the retaining washer (Fig. 7, item 1).
- Unscrew the hexagon screw (Fig. 7, item 2) at the motor shaft to lower the retaining washer (Fig. 7, item 1), thereby lowering the impeller/impeller shaft (Fig. 7, item 3). When the impeller has been completely lowered (Fig. 8, after a travel of about 5 mm), completely take out the hexagon screw and the retaining washer.
- Unscrew the coupling screw out of the installation drilled hole and remove the remaining coupling half (Fig. 9). If need be, use the provided press-off drilled holes.
- Unscrew the central screw (Fig. 10, item 2) of the impeller shaft and remove it with the retaining washer (Fig. 10, item 3).
- Take out both impeller shaft keys (Fig. 10, item 1).
- Carefully pull off and remove the mechanical seal (Fig. 11) from the impeller shaft.

Installation

Installation:



NOTE:

Thoroughly clean the sliding/seat surfaces of the impeller shaft and the lantern. If the shaft is damaged, it must be replaced. Always use new screws for the twist guard. Replace the O-rings in the cover groove and in the shaft sleeve groove with new ones.

- Completely screw in a cover screw (Fig. 12, item 1) into each of the two press-off drilled holes of the cover.
- Be sure that all hexagon socket screws (locking pins) are countersunk half way into the cover collar (Fig. 12).
- Put the mechanical seal onto the impeller shaft in such a way that the four drilled holes for the cover screws are opposite the threads (Fig. 13). **Attention:** If the drilled holes of the locking pins are not arranged at 90° to each other, pay attention to the installation position. The drilled holes must point towards the lantern window to facilitate accessibility of the grub screw (Fig. 2 or Fig. 3, depending on the pump type), push the mechanical seal on until the forcing screws are on the housing. A commercially available dishwashing liquid can be used as a lubricant.
- Check the keys in the motor shaft for correct seating.
- Push the motor shaft retaining washer on and fix with central screw (Fig. 14). Ensure that the motor shaft retaining washer is well seated when the central screw is completely screwed in and that the central screw thread bears at least 12 mm in the motor shaft thread in this position. In case necessary, use the supplied washers.
- By unscrewing the central screw, lower the retaining washer of the motor shaft by about 5 mm (Fig. 14).
- Put the first key (Fig. 15, item 1) in the impeller shaft, put on the retaining washer (Fig. 15, Pos. 2) of the impeller shaft and screw in the hexagon screw (Fig. 15, item 3) **by hand**.
- Rotate the motor shaft in such a way that the motor shaft key and the impeller shaft key oppose each other.
- Put the first coupling half on the two keys and the retaining washers (Fig. 16).

- Align the threaded hole in the impeller shaft retaining washer to the installation drilled hole of the coupling half.
- Insert one of the coupling screws into the installation drilled hole and screw in (Fig. 17) half way.



NOTE:

When tightening screw connections in conjunction with the work described in the following: Observe the screw tightening torque for the thread type (see list "Table 5: Screw tightening torques" on page 54).

- Tighten the central screw of the impeller shaft with the prescribed torque. Use a strap wrench to apply counter pressure.
- Tighten the coupling screw (Fig. 17).
- Tighten the central screw of the motor shaft with the prescribed torque (Fig. 18, item 1). Use a strap wrench to apply counter pressure.
- Insert the second key (Fig. 19, item 2) of the impeller shaft.
- Put on the second coupling half (Fig. 19, item 1).
- Evenly screw the available coupling screws, as a final step, the coupling screw from the installation drilled hole (Fig. 20).
- Unscrew both mechanical seal press-out forcing screws from the cover (Fig. 21).
- Screw in the 4 cover screws (Fig. 22, item 1) and tighten them with the prescribed torque.
- Rescrew and tighten the four hexagon screws (locking pin, Fig. 22, item 2) completely in succession.
- Install the coupling guard metal sheets (Fig. 23).
- Connect the motor cable.

9.2.3 Changing the motor

The motor bearings are maintenance-free. Increased bearing noises and unusual vibrations indicate bearing wear. The bearing or motor must then be changed. The drive may only be replaced by the Wilo customer service.

- Disconnect the system from the power and secure it against being switched on.
- Make sure it is not live.
- Earth the work area and short-circuit.
- Close the shut-off device upstream and downstream of the pump.
- Depressurise the pump by opening the venting screw (Fig. 29/30/31, item 1.31).

Dismantling

Dismantling:



DANGER! Risk of fatal injury!

There is a risk of fatal injury from electric shock when working on electrical equipment.

- **Before working on electrical equipment, switch it off and secure it against being switched on again.**



DANGER! Risk of scalding!

Due to high fluid temperatures there is a danger of scalding.

- **At high fluid temperatures, let the pump cool down before starting any work.**
- Disconnect the motor connection cables.
- Dismantle the coupling guard metal sheets (Fig. 1, item 1).
- Press the mechanical seal out of its seat and dismantle the coupling (see "Dismantling" section in chapter 9.2.2 "Replacing the mechanical seal" on page 50 and Fig. 1 ... 9).

**WARNING! Risk of injury!**

Incorrect dismantling of the motor can result in personal injury.

- **Before dismantling the motor, be sure that the centre of gravity is not above the holding point.**
- **Secure the motor against tipping over during transport.**
- **Always use suitable lifting equipment and secure parts against falling.**
- **Never stand underneath a suspended load.**
- Loosen the motor fastening screws (Fig. 25, item 1) at the motor flange (Fig. 26).
- Lift up the motor with suitable hoisting gear from the pump (Fig. 27).
- Install the new motor using suitable hoisting gear and diagonally tighten the connection between lantern and motor. (Fig. 28).

**NOTE:**

When tightening screw connections in conjunction with the work described in the following: Observe the screw tightening torque for the thread type (see list "Table 5: Screw tightening torques" on page 54).

- Check the coupling sliding surfaces and the shaft sliding surfaces; clean if required.
- Install the coupling and fasten the mechanical seal (see "Installation" section in chapter 9.2.2 "Replacing the mechanical seal" on page 50 and Fig. 14 ...22).
- Install the coupling guard metal sheets (Fig. 23).
- Connect the motor cable.

Screw tightening torques

Screw connection			Tightening torque Nm ± 10 %	Installation information
Post	Size/tensile strength class			
Impeller — shaft	M20	A2–70	100	Lubricate the thread with Molykote® P37 or comparable
	M18		145	
	M24		350	
Pump housing — Lantern	M16	8.8	100	Tighten evenly and diagonally
	M20		170	
Lantern — motor	M16		100	
	M20		170	
Coupling	M10	10.9	60	Tighten the screws uniformly, keep the gap even on both sides
	M12		100	
	M16		230	
IL 250: Baseplate — Pump housing	M20	8.8	170	
Mechanical seal — Shaft	M6		7	Rescrew and tighten every screw separately
Mechanical seal — Lantern	M8	8.8	25	
	M10	8.8	35	
	M10	8.8	35	
Retaining washer — Impeller shaft	M16	8.8	60	
Retaining washer — Motor shaft	M20	8.8	60	

Table 5: Screw tightening torques

10 Faults, causes and remedies

Have faults remedied by qualified personnel only! Observe the safety instructions in chapter 9 "Maintenance" on page 49.

- **If the malfunction cannot be rectified, consult a specialist technician or the nearest customer service centre or representative office.**

Fault	Cause	Remedy
Pump does not start or stops working	Pump blocked	Disconnect motor from power supply, remove cause of blockage; if motor blocked, overhaul/replace motor/motor impeller unit
	Incorrectly installed mechanical seal	Remove the mechanical seal; replace defective parts; install mechanical seal in accordance to the instructions
	Cable terminal loose	Check all cable connections
	Fuses defective	Check fuses; replace faulty fuses
	Motor damaged	Have the motor checked by Wilo customer service or a specialised service centre and serviced if necessary
	Motor protection switch has triggered	Throttle the pump to the rated volume flow on the pressure side
	Motor protection switch set incorrectly	Set the motor protection switch to the correct rated current as shown on the rating plate
	Motor protection switch affected by excessive ambient temperature	Move the motor protection switch or protect it using heat insulation
	PTC thermistor tripping unit has triggered	Check the motor and fan cover for contaminants and clean if necessary, check ambient temperature and ensure an ambient temperature of $\leq 40^\circ\text{C}$ by forced venting if necessary
Pump runs at reduced Power	Incorrect direction of rotation	Check direction of rotation, change if necessary
	Stop valve on the pressure side throttled	Slowly open the stop valve
	Speed too slow	Remedy incorrect terminal bridging (Y instead of Δ)
	Air in the inlet pipe	Repair leaks on flanges, vent the pump, change the mechanical seal in case of visible leakage
Pump making noises	Cavitation due to insufficient supply pressure	Increase supply pressure, observe minimum pressure at the suction port, check slide valve and filter on the suction side and clean if need be
	Incorrectly installed mechanical seal	Remove the mechanical seal; replace defective parts; install mechanical seal in accordance to the instructions
	Motor has bearing damage	Have the pump checked by Wilo customer service or a specialised service centre and repaired if necessary
	Impeller grinding	Check faces and centrings and between lanterns and motor and lantern and pump housing; clean if necessary. Check the coupling and shaft sliding surfaces. If necessary, clean and lightly oil them.

Table 6: Faults, causes and remedies

11 Spare parts

Spare parts can be ordered from your local specialist and/or via Wilo customer service.

To avoid queries and incorrect orders, all data from the rating plate of the pump and the motor must be specified with every order.



CAUTION! Risk of property damage!

Trouble-free pump operation can only be guaranteed when original spare parts are used.

- Use only genuine Wilo spare parts.
- Each component is identified in the table below.
Information to be provided when ordering spare parts:
 - Spare part number
 - Name/description of the spare part
 - All data on the pump and motor rating plate



NOTE:

For a list of genuine spare parts, see the Wilo spare parts documentation.

Spare parts table

For the allocation of assemblies, see Fig. 29/30/31.

No.	Part	Details	No.	Part	Details
1	Exchange kit (complete)		1.5	Coupling (complete)	
1.1	Impeller (kit) with:		2	Motor	
1.11		Nut	3	Pump housing (kit) with:	
1.12		Conical spring washer	1.14		O-ring
1.13		Impeller	3.1		Pump housing (IL, DL, BL)
1.14		O-ring	3.2		Stopper for pressure measurement connections
1.2	Mechanical seal (kit) with:		3.3		
1.11		Nut	3.5		
1.12		Conical spring washer	4	Fastening screws for lantern/pump housing	
1.14		O-ring	5	Fastening screws for motor/lantern	
1.21		Mechanical seal	6	Nut for motor/lantern fixation	
1.3	Lantern (kit) with:		7	Washer for motor/lantern fixation	
1.11		Nut	8	Adapter ring (only BL pumps)	
1.12		Conical spring washer			
1.14		O-ring			
1.31		Air vent valve			
1.32		Coupling guard			
1.33		Lantern			
1.4	Coupling/shaft (kit) with:				
1.11		Nut			
1.12		Conical spring washer			
1.14		O-ring			
1.41		Complete coupling/shaft			
1.42		Circlip			

Table 7: Spare parts table

12 Disposal

Proper disposal and recycling of this product prevents damage to the environment and risks to personal health.

Disposal in accordance with the regulations requires the product to be drained and cleaned.

Lubricants must be collected. The pump components are to be separated according to material (metal, plastic, electronics).

1. Use public or private disposal organisations when disposing of all or part of the product.
2. For more information on proper disposal, please contact your local council or waste disposal office or the supplier from whom you obtained the product.



NOTE:

The product or any of its parts must not be disposed of with household waste!

For further information on recycling, visit www.wilo-recycling.com

Subject to change without prior notice!



**EU/EG KONFORMITÄTSERKLÄRUNG
EU/EC DECLARATION OF CONFORMITY
DECLARATION DE CONFORMITE UE/CE**

Als Hersteller erklären wir hiermit, dass die Pumpenbauarten der Baureihen

We, the manufacturer, declare that the pump types of the series

Nous, fabricant, déclarons que les types de pompes des séries

**IL
DL
BL**

(Die Seriennummer ist auf dem Typenschild des Produktes nach Punkten b) & c) von §1.7.4.2 und §1.7.3 des Anhanges I der Maschinenrichtlinie angegeben. / The serial number is marked on the product site plate according to points b) & c) of §1.7.4.2 and §1.7.3 of the annex I of the Machinery directive. / Le numéro de série est inscrit sur la plaque signalétique du produit en accord avec les points b) & c) du §1.7.4.2 et du §1.7.3 de l'annexe I de la Directive Machines.)

in der gelieferten Ausführung folgenden einschlägigen Bestimmungen entsprechen :

In their delivered state comply with the following relevant directives :

dans leur état de livraison sont conformes aux dispositions des directives suivantes :

_ Maschinenrichtlinie 2006/42/EG

_ Machinery 2006/42/EC

_ Machines 2006/42/CE

und gemäß Anhang 1, §1.5.1, werden die Schutzziele der Niederspannungsrichtlinie 2014/35/EU ab 20 April 2016 eingehalten
and according to the annex 1, §1.5.1, comply with the safety objectives of the Low Voltage Directive 2014/35/EU from April 20th 2016
et, suivant l'annexe 1, §1.5.1, respectent les objectifs de sécurité de la Directive Basse Tension 2014/35/UE à partir du 20/04/2016

_ Elektromagnetische Verträglichkeit-Richtlinie 2014/30/EU ab 20 April 2016

_ Electromagnetic compatibility 2014/30/EU from April 20th 2016

_ Compatibilité électromagnétique 2014/30/UE à partir du 20 avril 2016

_ Richtlinie energieverbrauchsrelevanter Produkte 2009/125/EG

_ Energy-related products 2009/125/EC

_ Produits liés à l'énergie 2009/125/CE

Nach den Ökodesign-Anforderungen der Verordnung 640/2009 für Ausführungen mit einem einstufigen Dreiphasen - 50Hz - Käfigläufer - Induktionselektromotor, der Verordnung 4/2014 "Geänderte" / Nach den Ökodesign-Anforderungen der Verordnung 547/2012 für Wasserpumpen, This applies according to eco-design requirements of the regulation 640/2009 to the versions with an induction electric motor, squirrel cage, three-phase, single speed, running at 50Hz, amended by Regulation 4/2014 " / This applies according to eco-design requirements of the regulation suivant les exigences d'éco-conception du règlement 640/2009 aux versions comportant un moteur électrique à induction à cage d'écureuil, triphasé, mono-vitesse, fonctionnant à 50Hz, amendé par le règlement 4/2014" / suivant les exigences d'éco-conception du règlement 547/2012

und entsprechender nationaler Gesetzgebung,

and with the relevant national legislation,

et aux législations nationales les transposant,

sowie auch den Bestimmungen zu folgenden harmonisierten europäischen Normen :

comply also with the following relevant harmonized European standards :

sont également conformes aux dispositions des normes européennes harmonisées suivantes :

EN 809+A1

**EN 60034-1
EN 60204-1**

EN 60034-30-1

Bevollmächtigter für die Zusammenstellung der technischen Unterlagen ist:

Person authorized to compile the technical file is :

Personne autorisée à constituer le dossier technique est :

Dortmund,



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Senior Vice President - Group ITQ

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**EU/EG KONFORMITÄTSERKLÄRUNG
EU/EC DECLARATION OF CONFORMITY
DECLARATION DE CONFORMITE UE/CE**

Als Hersteller erklären wir hiermit, dass die Pumpenbauarten der Baureihen
We, the manufacturer, declare that the pump types of the series
Nous, fabricant, déclarons que les types de pompes des séries

BL80/...
BL100/...
BL125/...
BL150/...

(Die Seriennummer ist auf dem Typenschild des Produktes nach Punkten b) & c) von §1.7.4.2 und §1.7.3 des Anhangs I der Maschinenrichtlinie angegeben. / The serial number is marked on the product site plate according to points b) & c) of §1.7.4.2 and §1.7.3 of the annex I of the Machinery directive. / Le numéro de série est inscrit sur la plaque signalétique du produit en accord avec les points b) & c) du §1.7.4.2 et du §1.7.3 de l'annexe I de la Directive Machines.)

in der gelieferten Ausführung folgenden einschlägigen Bestimmungen entsprechen :
In their delivered state comply with the following relevant directives :
dans leur état de livraison sont conformes aux dispositions des directives suivantes :

– **Maschinenrichtlinie 2006/42/EG**

– **Machinery 2006/42/EC**

– **Machines 2006/42/CE**

und gemäss Anhang 1, §1.5.1, werden die Schutzziele der Niederspannungsrichtlinie 2014/35/EU ab 20 April 2016 eingehalten*
*and according to the annex 1, §1.5.1, comply with the safety objectives of the Low Voltage Directive 2014/35/EU from April 20th 2016 **
*et, suivant l'annexe 1, §1.5.1, respectent les objectifs de sécurité de la Directive Basse Tension 2014/35/UE à partir du 20/04/2016 **

– **Elektromagnetische Verträglichkeit-Richtlinie 2014/30/EU ab 20 April 2016***

– **Electromagnetic compatibility 2014/30/EU from April 20th 2016***

– **Compatibilité électromagnétique 2014/30/UE à partir du 20 avril 2016***

– **Richtlinie energieverbrauchsrelevanter Produkte 2009/125/EG**

– **Energy-related products 2009/125/EC**

– **Produits liés à l'énergie 2009/125/CE**

Nach den Ökodesign-Anforderungen der Verordnung 640/2009* für Ausführungen mit einem einstufigen Dreiphasen - 50Hz - Käfigläufer - Induktionselektromotor, der Verordnung 4/2014* Geänderte / Nach den Ökodesign-Anforderungen der Verordnung 547/2012 für Wasserpumpen, This applies according to eco-design requirements of the regulation 640/2009* to the versions with an induction electric motor, squirrel cage, three-phase, single speed, running at 50Hz, amended by Regulation 4/2014* / This applies according to eco-design requirements of the suivant les exigences d'éco-conception du règlement 640/2009* aux versions comportant un moteur électrique à induction à cage d'écureuil, triphasé, mono-vitesse, fonctionnant à 50Hz, amendé par le règlement 4/2014* / suivant les exigences d'éco-conception du règlement 547/2012

und entsprechender nationaler Gesetzgebung,
and with the relevant national legislation,
et aux législations nationales les transposant,

sowie auch den Bestimmungen zu folgenden harmonisierten europäischen Normen :
comply also with the following relevant harmonized European standards :
sont également conformes aux dispositions des normes européennes harmonisées suivantes :

EN 809+A1

EN 60034-1*
EN 60204-1*

EN 60034-30-1*

*

Nicht anwendbar auf Barshaft Ausführungen (Pumpen ohne Elektromotor)
Not applicable on Barshaft versions (pumps without electric motor)
Non applicable sur les versions Bareshaft (pompes sans moteur électrique)

Bevollmächtigter für die Zusammenstellung der technischen Unterlagen ist:

Person authorized to compile the technical file is :

Personne autorisée à constituer le dossier technique est :

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(BG) - български език ДЕКЛАРАЦИЯ ЗА СЪОТЕТСТВИЕ ЕО WILO SE декларира, че продуктите посочени в настоящата декларация съответстват на разпоредбите на следните европейски директиви и приелите ги национални законодателства: Машины 2006/42/ЕО ; Електромагнитна съвместимост 2014/30/ЕО ; Продукти, свързани с енергопотреблението 2009/125/ЕО както и на хармонизираните европейски стандарти, упоменати на предишната страница.	(CS) - Čeština ES PROHLÁŠENÍ O SHODĚ WILO SE prohlašuje, že výrobky uvedené v tomto prohlášení odpovídají ustanovením níže uvedených evropských směrnic a národním právním předpisům, které je přejímají: Stroje 2006/42/ES ; Elektromagnetická Kompatibilita 2014/30/ES ; Výrobků spojených se spotřebou energie 2009/125/ES a rovněž splňují požadavky harmonizovaných evropských norem uvedených na předcházející stránce.
(DA) - Dansk EF-OVERENSSTEMMELSESERKLÆRING WILO SE erklærer, at produkterne, som beskrives i denne erklæring, er i overensstemmelse med bestemmelserne i følgende europæiske direktiver, samt de nationale lovgivninger, der gennemfører dem: Maskiner 2006/42/EF ; Elektromagnetisk Kompatibilitet 2014/30/EF ; Energirelaterede produkter 2009/125/EF De er ligeledes i overensstemmelse med de harmoniserede europæiske standarder, der er anført på forrige side.	(EL) - Ελληνικά ΔΗΛΩΣΗ ΣΥΜΜΟΡΦΩΣΗΣ ΕΚ WILO SE δηλώνει ότι τα προϊόντα που ορίζονται στην παρούσα ευρωπαϊκά δηλωσή είναι σύμφωνα με τις διατάξεις των παρακάτω οδηγιών και τις εθνικές νομοθεσίες στις οποίες έχει μεταφερθεί: Μηχανήματα 2006/42/ΕΚ ; Ηλεκτρομαγνητικής συμβατότητας 2014/30/ΕΚ ; Συνδεδόμενα με την ενέργεια προϊόντα 2009/125/ΕΚ και επίσης με τα εξής εναρμονισμένα ευρωπαϊκά πρότυπα που αναφέρονται στην προηγούμενη σελίδα.
(ES) - Español DECLARACIÓN CE DE CONFORMIDAD WILO SE declara que los productos citados en la presenta declaración están conformes con las disposiciones de las siguientes directivas europeas y con las legislaciones nacionales que les son aplicables : Máquinas 2006/42/CE ; Compatibilidad Electromagnética 2014/30/CE ; Productos relacionados con la energía 2009/125/CE Y igualmente están conformes con las disposiciones de las normas europeas armonizadas citadas en la página anterior.	(ET) - Eesti keel EÜ VASTAVUSDEKLARATSIOONI WILO SE kinnitab, et selles vastavustunnistuses kirjeldatud tooted on kooskõlas alljärgnevat Euroopa direktiivide sätetega ning riiklike seadusandlustega, mis nimetatud direktiivid üle on võtnud: Masinaid 2006/42/EÜ ; Elektromagnetilist Ühilduvust 2014/30/EÜ ; Energiatõrjuga toodete 2009/125/EÜ Samuti on tooted kooskõlas eelmisel leheküljel ära toodud harmoniseeritud Euroopa standarditega.
(FI) - Suomen kieli EY-VAATIMUSTENMUKAISUUSVAKUUTUS WILO SE vakuuttaa, että tässä vakuutuksessa kuvatut tuotteet ovat seuraavien eurooppalaisten direktiivien määräysten sekä niihin sovellettavien kansallisten lakiasetusten mukaisia: Koneet 2006/42/EY ; Sähkömagneettinen Yhteensopivuus 2014/30/EY ; Energiaan liittyvien tuotteiden 2009/125/EY Lisäksi ne ovat seuraavien edellisellä sivulla mainittujen yhdenmukaistettujen eurooppalaisten normien mukaisia.	(GA) - Gaeilge EC DEARBHÚ COMHLÍONTA WILO SE ndearbhaíonn an cur síos ar na táirgí atá i ráiteas seo, siad i gcomhréir leis na forálacha atá sna treoracha seo a leanas na hEorpa agus leis na dlíthe náisiúnta is infheidhme orthu: Innealra 2006/42/EC ; Comhoiriúnacht Leictreamaighnéadach 2014/30/EC ; Fuinneamh a bhaineann le táirgí 2009/125/EC Agus siad i gcomhréir le forálacha na caighdeáin chomhchuibhithe na hEorpa dá dtagraítear sa leathanach roimhe seo.
(HR) - Hrvatski EZ IZJAVA O SUKLADNOSTI WILO SE izjavlja da su proizvodi navedeni u ovoj izjavi u skladu sa sljedećim prihvaćenim europskim direktivama i nacionalnim zakonima: EZ smjernica o strojevima 2006/42/EZ ; Elektromagnetna kompatibilnost - smjernica 2014/30/EZ ; Smjernica za proizvode relevantne u pogledu potrošnje energije 2009/125/EZ i usklađenim europskim normama navedenim na prethodnoj stranici.	(HU) - Magyar EK-MEGFELELŐSÉGI NYILATKOZAT WILO SE kijelenti, hogy a jelen megfelelősségi nyilatkozatban megjelölt termékek megfelelnek a következő európai irányelvek előírásainak, valamint azok nemzeti jogrendbe áttültetett rendelkezéseinek: Gépek 2006/42/EK ; Elektromágneses összeférhetőségre 2014/30/EK ; Energiával kapcsolatos termékek 2009/125/EK valamint az előző oldalon szereplő, harmonizált európai szabványoknak.
(IS) - Íslenska EB LEYFISYFIRLÝSING WILO SE lýsir því yfir að vörurnar sem um getur í þessari yfirlýsingu eru í samræmi við eftirfarandi tilskipunum ESB og landslögum hafa samþykkt: Vélartilskipun 2006/42/EB ; Rafseguls-samhæfni-tilskipun 2014/30/EB ; Tilskipun varðandi vörur tengdar orkunotkun 2009/125/EB og samhæfða evrópska staðla sem nefnd eru í fyrri síðu.	(IT) - Italiano DICHIARAZIONE CE DI CONFORMITÀ WILO SE dichiara che i prodotti descritti nella presente dichiarazione sono conformi alle disposizioni delle seguenti direttive europee nonché alle legislazioni nazionali che le traspongono : Macchine 2006/42/CE ; Compatibilità Elettromagnetica 2014/30/CE ; Prodotti connessi all'energia 2009/125/CE E sono pure conformi alle disposizioni delle norme europee armonizzate citate a pagina precedente.
(LT) - Lietuvių kalba EB ATITIKTIES DEKLARACIJA WILO SE pareiškia, kad šioje deklaracijoje nurodyti gaminiai atitinka šių Europos direktyvų ir jas perkeliančių nacionalinių įstatymų nuostatus: Mašinos 2006/42/EB ; Elektromagnetinis Suderinamumas 2014/30/EB ; Energija susijusiems gaminiams 2009/125/EB ir taip pat harmonizuotas Europas normas, kurios buvo cituotos ankstesniame puslapyje.	(LV) - Latviešu valoda EK ATBILSTĪBAS DEKLARĀCIJU WILO SEdeklarē, ka izstrādājumi, kas ir nosaukti šajā deklarācijā, atbilst šeit uzskaitīto Eiropas direktīvu nosacījumiem, kā arī atsevišķu valstu likumiem, kuros tie ir ietverti: Mašīnas 2006/42/EK ; Elektromagnētiskās Saderības 2014/30/EK ; Energiju saistītiem ražojumiem 2009/125/EK un saskaņotajiem Eiropas standartiem, kas minēti iepriekšējā lappusē.

(MT) - Malti DIKJARAZZJONI KE TA' KONFORMITÀ WILO SE jiddikjara li l-prodotti speċifikati f'din id-dikjarazzjoni huma konformi mad-direttivi Ewropej li jsegwu u mal-leġislazzjonijiet nazzjonali li japplikawhom: Makkinarju 2006/42/KE ; Kompatibbiltà Elettromanjetika 2014/30/KE ; Prodotti relatati mal-enerġija 2009/125/KE kif ukoll man-normi Ewropej armonizzati li jsegwu imsemmija fil-paġna preċedenti.	(NL) - Nederlands EG-VERKLARING VAN OVEREENSTEMMING WILO SE verklaart dat de in deze verklaring vermelde producten voldoen aan de bepalingen van de volgende Europese richtlijnen evenals aan de nationale wetgevingen waarin deze bepalingen zijn overgenomen: Machines 2006/42/EG ; Elektromagnetische Compatibiliteit 2014/30/EG ; Energiegerelateerde producten 2009/125/EG De producten voldoen eveneens aan de geharmoniseerde Europese normen die op de vorige pagina worden genoemd.
(NO) - Norsk EU-OVERENSSTEMMELSESERKLÆING WILO SE erklærer at produktene nevnt i denne erklæringen er i samsvar med følgende europeiske direktiver og nasjonale lover: EG-Maskindirektiv 2006/42/EG ; EG-EMV-Elektromagnetisk kompatibilitet 2014/30/EG ; Direktiv energirelaterte produkter 2009/125/EF og harmoniserte europeiske standarder nevnt på forrige side.	(PL) - Polski DEKLARACJA ZGODNOŚCI WE WILO SE oświadcza, że produkty wymienione w niniejszej deklaracji są zgodne z postanowieniami następujących dyrektyw europejskich i transponującymi je przepisami prawa krajowego: Maszyn 2006/42/WE ; Kompatybilności Elektromagnetycznej 2014/30/WE ; Produktów związanych z energią 2009/125/WE oraz z następującymi normami europejskich zharmonizowanymi podanymi na poprzedniej stronie.
(PT) - Português DECLARAÇÃO CE DE CONFORMIDADE WILO SE declara que os materiais designados na presente declaração obedecem às disposições das directivas europeias e às legislações nacionais que as transcrevem : Máquinas 2006/42/CE ; Compatibilidade Electromagnética 2014/30/CE ; Produtos relacionados com o consumo de energia 2009/125/CE E obedecem também às normas europeias harmonizadas citadas na página precedente.	(RO) - Română DECLARAȚIE DE CONFORMITATE CE WILO SE declară că produsele citate în prezenta declarație sunt conforme cu dispozițiile directivelor europene următoare și cu legislațiile naționale care le transpun : Mașini 2006/42/CE ; Compatibilitate Electromagnetică 2014/30/CE ; Produselor cu impact energetic 2009/125/CE și, de asemenea, sunt conforme cu normele europene armonizate citate în pagina precedentă.
(RU) - русский язык Декларация о соответствии Европейским нормам WILO SE заявляет, что продукты, перечисленные в данной декларации о соответствии, отвечают следующим европейским директивам и национальным предписаниям: Директива ЕС по машинному оборудованию 2006/42/ЕС ; Директива ЕС по электромагнитной совместимости 2014/30/ЕС ; Директива о продукции, связанной с энергопотреблением 2009/125/ЕС и гармонизированным европейским стандартам, упомянутым на предыдущей странице.	(SK) - Slovenčina ES VYHLÁSENIE O ZHODE WILO SE čestne prehlasuje, že výrobky ktoré sú predmetom tejto deklarácie, sú v súlade s požiadavkami nasledujúcich európskych direktív a odpovedajúcich národných legislatívnych predpisov: Strojových zariadeniach 2006/42/ES ; Elektromagnetickú Kompatibilitu 2014/30/ES ; Energeticky významných výrobkov 2009/125/ES ako aj s harmonizovanými európskych normami uvedenými na predchádzajúcej strane.
(SL) - Slovenščina ES-IZJAVA O SKLADNOSTI WILO SE izjavlja, da so izdelki, navedeni v tej izjavi, v skladu z določili naslednjih evropskih direktiv in z nacionalnimi zakonodajami, ki jih vsebujejo: Stroji 2006/42/ES ; Elektromagnetno Združljivostjo 2014/30/ES ; Izdelkov, povezanih z energijo 2009/125/ES pa tudi z usklajenimi evropskih standardi, navedenimi na prejšnji strani.	(SV) - Svenska EG-FÖRSÄKRAN OM ÖVERENSSTÄMMELSE WILO SE intygat att materialet som beskrivs i följande intyg överensstämmer med bestämmelserna i följande europeiska direktiv och nationella lagstiftningar som inför dem: Maskiner 2006/42/EG ; Elektromagnetisk Kompatibilitet 2014/30/EG ; Energirelaterade produkter 2009/125/EG Det överensstämmer även med följande harmoniserade europeiska standarder som nämnts på den föregående sidan.
(TR) - Türkçe CE UYGUNLUK TEYID BELGESİ WILO SEbu belgede belirtilen ürünlerin aşağıdaki Avrupa yönetmeliklerine ve ulusal kanunlara uygun olduğunu beyan etmektedir: Makine Yönetmeliği 2006/42/AT ; Elektromanyetik Uyumluluk Yönetmeliği 2014/30/AT ; Eko Tasarım Yönetmeliği 2009/125/AT ve önceki sayfada belirtilen uyumlaştırılmış Avrupa standartlarına.	

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